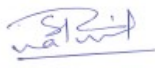


	FIR VEHICLE TESTING		20/05/2025
FIR No :FIR/Testing-Outdoor-UA/520700/2025/00412			
Part No/ Group :52075256R			
Subject :Performance evaluation of Signa 3523.T BSVI-II AC ON in comparison with Signa 3523.T BSVI-II AC OFF vehicle			
Test Vehicle	Project Name :Signa 3523.T BSVI-II AC Regulation		Enclosures : Annexure 1-5
	Model Name : Signa 3523.T BSVI-II		
	Chasis Number :PC53562032RJ00873		
	Engine Number :22L63919785		
FIR Status :	ESO		

Role	Name	Signature
Prepared By	Palash Das	 20-May-2025
Checked By	Swapnil Jadhav	 20-May-2025
Approved By	Ajaykumar Jadhav	 20-May-2025

Cc: Head CVBU Engg, Chief Engineer, Head COC, Project Team [Design]; Project Team [Testing]

PA 20 - F 222 / May19

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Documents : Please click here

[Signa_3523.T_.pdf](#)

 FIR/Testing-Outdoor-UA/520700/2025/00412	FIR VEHICLE TESTING	Date: 20/05/2025	
Part No / Group: 52075256R	Specification no :- " 259052070002 "	Action by	Failure class

Subject: -	Performance evaluation of Signa 3523.T BSVI-II AC ON in comparison with Signa 3523.T BSVI-II AC OFF vehicle.		
<u>Conclusion:</u> ➤ <u>Signa 3523.T BSVI-II AC ON in comparison with Signa 3523.T BSVI-II AC OFF.</u> ➤ Overall performance of Signa 3523.T BSVI-II is better as compared to Signa 3523.T BSVI-II AC OFF vehicle. 1.1 <u>Acceleration:</u> ➤ <u>Signa 3523.T BSVI-II AC ON in comparison with Signa 3523.T BSVI-II AC OFF:</u> With AC ON : ➤ Acceleration (0-60 kmph) of Signa 3523.T BSVI-II was found to be 60.88 Sec. Signa 3523.T BSVI-II is 0.95% better as compared to target. ➤ Acceleration (0-1000m) of Signa 3523.T BSVI-II was found to be 80.53 Sec. Signa 3523.T BSVI-II is 1.16% better as compared to target. With AC OFF: ➤ Acceleration (0-60 kmph) of Signa 3523.T BSVI-II was found to be 58.57 Sec. ➤ Acceleration (0-1000m) of Signa 3523.T BSVI-II was found to be 79.61 Sec. 1.2 <u>Wide open throttle drivability:</u> ➤ <u>Signa 3523.T BSVI-II AC ON in comparison with Signa 3523.T BSVI-II AC OFF:</u> - Rate of Acceleration of Signa 3523.T BSVI-II vehicle was found to be 1.87 kmph/s, 1.77 kmph/s, 1.44 kmph/s, 0.96 kmph/s, 0.61 kmph/s, 0.46 kmph/s, in 1st , 2nd , 3rd , 4th , 5th 6th, gear respectively.			
FIR Status		Page 1 of 4	

 FIR/Testing-Outdoor-UA/520700/2025/00412	FIR VEHICLE TESTING	Date: 20/05/2025	
Part No / Group: 52075256R	Specification no :- " 259052070002 "	Action by	Failure class

1.3 Bathtub drivability:

➤ Signa 3523.T BSVI-II AC ON in comparison with Signa 3523.T BSVI-II AC OFF:

Bathtub drivability of Signa 3523.T BSVI-II vehicle was found better as compared to targets.

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1.4 Sustain and Restart Gradability:

➤ Signa 3523.T BSVI-II AC ON in comparison with Signa 3523.T BSVI-II AC OFF:

- Sustain gradeability of Signa 3523.T BSVI-II was carried forward from Signa 3523.T BSVI-II AC OFF and is meeting the PALS target.
- Restart gradeability of Signa 3523.T BSVI-II was carried forward from Signa 3523.T BSVI-II AC OFF and is meeting the PALS target.
- Note:-
- Target has been derived from delta of AC ON & OFF Simulation data of Signa 3523.T BSVI-II & applied to actual tested data of Signa 3523.T BSVI-II AC OFF.
- As Per worse case matrix, Signa 3523.T BSVI-II vehicle validated & AC performance delta (Drop) extended to following vehicle models (Refer Annexure No- 6)

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Extension Matrix Models				
SI No	Vehicle PAT	Engine	Lead vehicle/Vehicle Tested	Variant
1	PD	5.6L (230HP)	SIGNA/LPT 3523.T	SIGNA/LPT 3723.T
2				SIGNA/LPT 3523.T
3				SIGNA/LPT 3123.T
4				SIGNA/LPT 3023.T
5				SIGNA/LPT 2823.T
6				SIGNA/LPT 2623.T

2.0 Result: Wide observation.

FIR Status

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 FIR/Testing-Outdoor-UA/520700/2025/00412	FIR VEHICLE TESTING	Date: 20/05/2025
Part No / Group: 52075256R	Specification no :- " 259052070002 "	Action by Failure class

FAILURES CLASS:

- A: Failure that happen without warning. Detrimental to vehicle / personnel, road users and safety. Can cause death / hospitalization. Noncompliance of regulatory requirement.
- B: Vehicle gets crippled with sufficient warning. Minor injury to occupant & road users (First aid only). Premature failures / repeated failures. Leads to consequential catastrophic damages. Need immediate attention.
- C: Customer irritants. Capable of damaging aesthetics. Failure can be attended at next servicing Schedule.
- D: Customer perception issues and minor irritants.

3.0 Test Vehicle:

- Signa 3523.T BSVI-II (ERC-1309)
Chassis No. – PC53562032RJ00873
Engine No. – 22L63919785

Enclosures:

- Annexure 1: Panel chart
Annexure 2: PAT Sheet
Annexure 3: PAT Summary
Annexure 4: Test request
Annexure 5: Test plan
Annexure 6: Mail Comm for Extension matrix & Cal ID
Annexure 7: Cal ID

Input Doc Ref. : Testing-Outdoor-UA/01327

DID No : NA

DDP No : NA

Work Order No. :- P.23.EA.533

Make :- Tata Motors Ltd

T I No. – UA001181

Earlier Ref: - NA

4.0 Project Status: DR3
5.0 Introduction:

As a part of AC regulation program, Signa 3523.T BSVI-II vehicle was offered for AC ON PD evaluation, This report will provide a comprehensive overview of the trial's details and current status.

6.0 Testing: Following test was conducted during trials.

Sr. No.	Test Description	Standard Referred	Instrument used
1	Acceleration	TST_VT_WG_0002	VBOX SX with GPS, Weighbridge (CRS Gate)
2	WOT Drivability	TST_VT_WG_0002	
3	Bath -Tub Drivability	TST_VT_WG_0261	
4	Gradability	TST_VT_WG_11	
5	Restart Gradability	TST_VT_WG_127	
6	Coastdown	TST_VT_WG_36	

FIR Status

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 FIR/Testing-Outdoor-UA/520700/2025/00412	FIR VEHICLE TESTING	Date: 20/05/2025	
Part No / Group: 52075256R	Specification no :- " 259052070002 "	Action by	Failure class

7.0 Specification:

Parameter		Test vehicle
Vehicle model		Signa 3523.T BSVI-II (ERC-1309)
Engine Model		Cummins ISBe 5.6L/ Ashwamedh
Calibration ID		PF10040
CAL Used in Vehicle		XWTAA19D07
Max Power		220 HP @2300 RPM
Max Torque		925 Nm @ 1000 ~1300 rpm
GVW (kg)		35000
Gear Box		TATA G-950 DD
Gear Ratios.		1-9.36, 2- 5.20, 3- 3.05, 4- 1.84, 5- 1.33, 6- 1.00
Axle Ratio		5.12
Tyres	FA1	Size : 295/90R 20, Make: M/s Bridgestone, Type : Phase-II
	RA1	Size : 295/90R 20, Make: M/s Bridgestone, Type : Phase-II
	RA2	Size : 295/90R 20, Make: M/s Bridgestone, Type : Phase-II
Fan Details (Type/Make/Blades)		Type: E-Viscous fan and Size:24"

8.0 Torque table:-

➤ Balance mode (TF – 130%) – Test Vehicle

	600	800	1000	1200	1400	1600	1800	2000	2200	2400	2600	2800
1	500	500	850	753	400	400	400	400	400	400	80	0
1.33	500	500	850	800	753	543	400	400	400	400	80	0
1.84	500	500	850	800	753	543	400	400	400	400	80	0
3.05	500	500	850	800	800	753	543	400	400	400	80	0
5.2	450	500	850	850	850	850	772	694	616	412	80	0
9.36	450	500	850	850	850	850	772	694	616	412	80	0

➤ Eco mode (TF – 120%) – Test Vehicle

	600	800	1000	1200	1400	1600	1800	2000	2200	2400	2600	2800
1	500	500	850	753	400	400	400	400	400	400	80	0
1.33	500	500	850	800	753	543	400	400	400	400	80	0
1.84	500	500	850	800	753	543	400	400	400	400	80	0
3.05	500	500	850	800	800	753	543	400	400	400	80	0
5.2	450	500	850	850	850	850	772	694	616	412	80	0
9.36	450	500	850	850	850	850	772	694	616	412	80	0

9.0 Observation:- Refer annexure 7 for calibration ID mail of all other platforms

Vehicle PAT	Engine	Lead vehicle / Vehicle Tested	Truck Models	Engine Cal Id
PD	5.6L (230HP)	SIGNA/LPT 3523.T	SIGNA/LPT 3723.T	PF10040
			SIGNA/LPT 3523.T	PF10040
			SIGNA/LPT 3123.T	PF10040
			SIGNA/LPT 3023.T	PF10040
			SIGNA/LPT 2823.T	PF10040/PF10046
			SIGNA/LPT 2623.T	PF10040

10.0 Future testing: - As per future plan.

FIR Status	Page 4 of 4
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Level 1 Attribute:		PANEL chart		VE CVBU/ PD/001	
PAT- Performance and Drivability		Specification: Max Power-220 hp @ 2300 rpm Max Torque - 925 Nm 1000-1300 rpm Driveline : G-950 DD 5.12 RAR 295/90R20 Tyres iUSFE Torque Factor: E-130%, B-120% EOL Parameters : RSG: 80 kmph		Panel Chart No.: VE CVBU/ M&HCV/ Signa 3523.T BSVI-II/ PD/01	
Programme / MY : Signa 3523.T BSVI-II	Gateway : DR3	Owner: Ajaykumar Jadhav	Project Name: Signa 3523.T BSVI-II AC ON	Date: 16/05/2025	

Level 1 Summary:		Statement Of Intent (SOI) or DNA Description:			
PALS	Level 1:	PALS	Status		Performance: Haulage application
		Current/ Gateway	Current/ Gateway	Job#1 Predicted	
		C	--		

Attribute Target Summary:							Current Target	Status	Base Vehicle	Competitor data BSVI		
Level 2	Level 3	Level 4	Level 5	Level 6	PALS	Metric	Extrapolated	Current/Gateway Signa 3523.T BSVI-II	Signa 3523.T BSVI-II	Core Benchmark	Adjacent Benchmark	Remarks
								AC ON	AC OFF			
Specification								Max Power-220 hp @ 2300 rpm Max Torque - 925 Nm 1000-1300 rpm Driveline : G-950 DD 5.12 RAR 295/90R20 Tyres iUSFE Torque Factor: E-130%, B-120%	Max Power-220 hp @ 2300 rpm Max Torque - 925 Nm 1000-1300 rpm Driveline : G-950 DD 5.12 RAR 295/90R20 Tyres iUSFE Torque Factor: E-130%, B-120%	--	--	--
Vehicle GVW (Payload)						kg		35000	35000	--	--	--
Power (Max) @ rpm						HP		220 hp @ 2300 rpm	220 hp @ 2300 rpm	--	--	
Power to Weight ratio						HP / Ton		6.29	6.29	--	--	
Torque (Max) @ rpm						Nm		925 Nm @ 1000-1300 rpm	925 Nm @ 1000-1300 rpm	--	--	
Torque to Weight ratio						Nm / Ton		26.43	26.43	--	--	
Tyres size						~		Bridgestone 295/90R20	Bridgestone 295/90R20	--	--	--
Gear Box Ratios						~		G-950 DD, FGR -9.36	G-950 DD, FGR -9.36	--	--	
Final Drive Ratio						~		5.12	5.12	--	--	--
Acceleration at rated load					C			--		--	--	
0-60 kmph						Sec	≤ 61.46	60.88	58.57	--	--	
0-1000 m						Sec	≤ 81.48	80.53	79.61	--	--	
Bathtub Drivability (Time) at Rated load					C			--		--	--	
										--	--	Drivability of Signa 3523.T BSVI-II was found to be better as compared to Targets in all gears.
3rd Gear										--	--	
10-15 kmph						sec	≤ 4.62	4.53	4.62	--	--	
15-20 kmph						sec	≤ 3.26	3.25	3.26	--	--	
4th Gear										--	--	
20-25 kmph						sec	≤ 6.35	6.15	5.80	--	--	
25-30 kmph						sec	≤ 4.04	4.02	3.84	--	--	
30-35 kmph						sec	≤ 4.21	4.11	3.96	--	--	1. Signa 3523.T BSVI-II targets has been derived by Signa 3523.T BSVI-II AC ON & OFF Simulation delta applied to actual tested data of Signa 3523.T BSVI-II AC OFF data. 2. Signa 3523.T BSVI-II was having 100% engine calibration maturation. 3. AC PALS position to be carried forward from existing Non AC PALS position..
5th Gear										--	--	
25-30 kmph						sec	≤ 10.44	10.18	9.81	--	--	
30-35 kmph						sec	≤ 6.78	6.74	6.99	--	--	
35-40 kmph						sec	≤ 5.85	5.47	5.44	--	--	
										--	--	
6th Gear										--	--	
30-40 kmph						sec	≤ 30.45	30.35	29.43	--	--	
40-50 kmph						sec	≤ 17.42	17.36	16.49	--	--	
50-60 kmph						sec	≤ 18.93	17.83	16.98	--	--	
Gradeability at rated load					C			--		--	--	Sustain & Restart Gradeability is carried forward from Signa 3523.T AC OFF
Sustain Gradeability						%	≥ 24.76	25.37		--	--	
Restart Gradeability						%	≥ 18.34	18.80		--	--	

Executive Summary:	1. Signa 3523.T BSVI-II targets has been derived by Signa 3523.T BSVI-II AC ON & OFF Simulation delta applied to actual tested data of Signa 3523.T BSVI-II AC OFF data. 2. Signa 3523.T BSVI-II was having 100% engine calibration maturation. 3. AC PALS position to be carried forward from existing Non AC PALS position..
--------------------	--

Tested by:						Verified and Approved by:					
  Palash Das / Swapnil Jadhav PAT Engineer						  Ajaykumar Jadhav / Sanjeev Kumar PAT Lead / Project Lead					

TATA TATA MOTORS LTD. ERC,PUNE		PAT ATTRIBUTES SHEET Performance and Drivability Vehicle-level Engine Calibration Features / End of Line OBDII and Diagnostics										Document Version: VE CVBU/PD/001														
		Department Name: Vehicle Engineering										PAT Sheet No: VE CVBU/ HCV/ Signa 3523.T BSVI-II 001														
												Project Name: Signa 3523.T BSVI-II Max Power-220 hp @ 2300 rpm Max Torque - 925 Nm 1000-1300 rpm Driveline : G-950 DD 5.12 RAR 295/90R20 Tyres iUSFE Torque Factor: E-130%, B-120% Base Vehicle- Signa 3523.T BSVI-II														
Date: 16/05/2025																										
Sr.No.	Aggregate and test parameter						Unit	Test Type	Classification @QR3	Ref.Standard / Work Guideline	Simulation	Base Vehicle Signa 3523.T BSVI-II AC ON (Extrapolated from AC OFF)	Target	Test Vehicle Signa 3523.T BSVI-II With AC ON (G-950 DD 5.12 RAR 295/90R20)	Status (RYG)	Remarks	CoC Remark									
A	Performance Evaluation								Heading			--	--	Max Power-220 hp @ 2300 rpm Max Torque - 925 Nm 1000-1300 rpm Driveline : G-950 DD 5.12 RAR 295/90R20 Tyres iUSFE Torque Factor: E-130%, B-120%	--	--										
1.1	Minimum speed								Heading			Heavy Mode		Heavy Mode	--											
1.1.1	1st gear						Kmph		Specification			--		2.4	--	--										
1.1.2	2nd gear						Kmph		Specification			--		4.6	--	--										
1.1.3	3rd gear						Kmph		Specification			--		8.3	--	--										
1.1.4	4th gear						Kmph	Standard	Specification	TSTV/TWGO2		--	--	12.8	--	--										
1.1.5	5th gear						Kmph		Specification			--		18.0	--	--										
1.1.6	6th gear						Kmph		Specification			--		25.6	--	--										
1.2	Maximum speed								Heading						--											
1.2.1	1st gear						Kmph		Specification			--		8.0	--	--										
1.2.2	2nd gear						Kmph		Specification			--		14.5	--	--										
1.2.3	3rd gear						Kmph	Standard	Specification	TSTV/TWGO2		--	--	26.6	--	--										
1.2.4	4th gear						Kmph		Specification			--	--	46.7	--	--										
1.2.5	5th gear						Kmph		Specification			--	--	49.7	--	--										
1.2.6	6th gear						Kmph		Specification			--	--	75.0	--	--										
1.2.7	Vehicle maximum speed						Kmph		Target			80.0	≥ 80	80	Y	--										
1.3	Acceleration								Heading						--											
1.4.2	AC ON								Heading						--											
1.4.2.1	0-60 kmph (0-37 mph)						Sec	Standard	Target	TSTV/TWGO2		61.46	≤ 61.46	60.88	Y											
1.4.2.4	0-1000 m						Sec		Target			81.48	≤ 81.48	80.53	Y											
1.5	Grad ability								Heading						--											
1.5.2	AC ON								Heading						--											
1.5.2.1	Maximum (sustained) Gradability (forward)						%	Standard	Target	TSTV/TWGO11		24.76	≥ 24.76	25.37	Y	Sustain & Restart Gradability is carried forward from Signa 3523.T										
1.5.2.2	Max Restart Gradability (forward)						%	Standard	Target	TSTV/TWGO127		18.34	≥ 18.34	18.80	Y	AC OFF										
B	Drivability Evaluation								Heading					--												
1.3.1	WOT Drivability (AC ON)								Heading						--											
1.3.1.1	1st gear 4-12 Km/h						Sec		Data Generation			--	--	1.87	--											
1.3.2	2nd gear 5-18 Km/h						Sec		Data Generation			--	--	2.27	--											
1.3.3	3rd gear 7- 24 Km/h						Sec		Data Generation			--	--	5.57	--											
1.3.4	4th gear 10- 30 Km/h						Sec		Data Generation			--	--	14.56	--											
1.3.5	5th gear 14- 45 Km/h						Sec		Data Generation			--	--	24.39	--											
1.3.6	6th gear 20- 60 km/h						Sec		Data Generation			--	--	30.25	--											
1.3.10	AC ON (Rate of acceleration)								Heading	TSTV/TWGO2		--	--													
1.3.11.1	1st gear 4-12 Km/h						Kmph/sec	Standard	Data Generation			--	--	1.87	--											
1.3.12	2nd gear 5-18 Km/h						Kmph/sec		Data Generation			--	--	1.77	--											
1.3.13	3rd gear 7- 24 Km/h						Kmph/sec		Data Generation			--	--	1.44	--											
1.3.14	4th gear 10- 30 Km/h						Kmph/sec		Data Generation			--	--	0.96	--											
1.3.15	5th gear 14- 45 Km/h						Kmph/sec		Data Generation			--	--	0.63	--											
1.3.16	6th gear 20- 60 km/h						Kmph/sec		Data Generation			--	--	0.46	--											
1.4.7	Ballhub Curve - Part Throttle Drivability (AC ON)								Heading						--											
1.4.7.1	3rd gear								Heading						--											
1.4.11	10- 15 Km/h						sec		Target			4.62	≤ 4.62	4.53	Y											
1.4.12	15- 20 Km/h						sec		Target			3.26	≤ 3.26	3.25	Y											
1.4.13	4th gear								Heading						--											
1.4.14	20- 25 Km/h						sec		Target			6.26	≤ 6.26	6.28	Y											
1.4.15	25- 30 Km/h						sec		Target			4.04	≤ 4.04	4.02	Y											
1.4.16	30- 35 Km/h						sec		Target			4.21	≤ 4.21	4.31	Y											
1.4.1	5th gear								Heading	TSTV/TWGO1					--											
1.4.4	25- 30 Km/h						sec	Standard	Target			10.44	≤ 10.44	10.38	Y	Drivability of Signa 3523.T BSVI-II was found to be better as compared to Targets in all gears..										
1.4.6	30- 35 Km/h						sec		Target			6.78	≤ 6.78	6.74	Y											
1.4.6	35- 40 Km/h						sec		Target			9.85	≤ 9.85	9.47	Y											
1.4.8	6th gear								Heading						--											
1.4.9	30-40 km/h						sec		Target			30.45	≤ 30.45	30.35	Y											
1.4.10	40-50 km/h						sec		Target			17.42	≤ 17.42	17.35	Y											
1.4.11	50-60 km/h						sec		Target			18.93	≤ 18.93	17.83	Y											
C	Engine Calibration Activities								Heading					--												
1.0	System Verification Trials								Heading						--											
1.1	Powertrain Characteristics (AC ON)								Target				Pass	Pass	Y											
2.0	Vehicle Performance Trials								Heading						--											
2.1	Vehicle Speed detection								Target				Pass	Pass	Y											
2.2	Engine Speed detection								Target				Pass	Pass	Y											
2.3	Gear Detection								Target				Pass	Pass	Y											
2.4	Driven life								Target				Pass	Pass	Y											
3.0	Engine derate verification								Heading						--											
3.2	Limp home validation								Heading						--											
3.2.6	Limp home Engine speed that satisfies 7 deg Grade-ability								Target				Pass	Pass	Y	--										
3.2.7	Limp home Engine speed that satisfies 7 deg Restart Grade-ability								Target				Pass	Pass	Y	--										
3.2.8	Limp home Engine speed that satisfies both 7 deg Grade-ability and Restart Grade-ability								Target				Pass	Pass	Y	--										
3.2.9	% Deterioration in Drivability with Limp/home condition								Target				Pass	Pass	Y	--										
3.3	Reduced torque fault validation								Heading						--											
3.3.5	% Torque reduction that satisfies 7 deg Grade-ability								Target				Pass	Pass	Y	--										
3.3.6	% Torque reduction that satisfies 7 deg Restart Grade-ability								Target				Pass	Pass	Y	--										
3.3.7	% Torque reduction that satisfies both 7 deg Grade-ability and Restart Grade-ability								Target				Pass	Pass	Y	--										
3.3.8	% Deterioration in Drivability with Reduced torque fault condition								Target				Pass	Pass	Y	--										
4.0	Evaluation of Take-off ability, Acceleration and Deceleration								Heading						--											
4.1	Take-off ability- Subjective characterisation								Target				Pass	Pass	Y	--										
4.4	Acceleration performance (Subjective evaluation)								Target				Pass	Pass	Y	--										
D	Vehicle feature validation and Fuel Economy features / End of Line (EOL) parameters segregation and verification								Heading					--												
6.0	Vehicle acceleration management (VAM tuning)								Target				Pass	Pass	Y	--										
6.0	Electronic vehicle speed Limiter (EVL) / Load based speed control (LBSC)								Target				Pass	Pass	Y	--										
6.0	EVL+ VAM / LBSC+VAM								Target				Pass	Pass	Y	--										
7.0	Gear Down Protection (GDP)								Target				Pass	Pass	Y	--										
8.0	Idle shutdown								Target				Pass	Pass	Y	--										
9.0	Average fuel economy (AFE)								Target				Pass	Pass	Y	--										
10.0	Engine Brake control								Target				Pass	Pass	Y	--										
10.0	Cruise control (CC)								Target				Pass	Pass	Y	--										
10.0	Road Speed governor								Target				Pass	Pass	Y	--										
10.0	Trip-info								Target				Pass	Pass	Y	--										
E	Environmental Trials Verification								Heading					--												
1.0	Summer trials								Heading						--											
1.4	Hot start								Target				Pass	Pass	Y	--										
1.5	Low idle governor calibration								Target				Pass	Pass	Y	--										
1.6	Driven life								Target				Pass	Pass	Y	--										
1.7	Subjective Drivability Assessment								Target				Pass	Pass	Y	--										
1.7.1	Jerks observed during acceleration and deceleration in unladen condition or not.								Target				Pass	Pass	Y	--										
1.7.2	Jerks observed during acceleration and deceleration in laden condition or not.								Target				Pass	Pass	Y	--										
2.0	Winter trials								Heading						--											
2.2	Cold Start (as Ambient as per Application requirement)								Target				Pass	Pass	Y	--										
2.3	Smoke level (visually) during free acceleration after start								Target				Pass	Pass	Y	--										
2.3	Subjective Drivability Assessment								Target				Pass	Pass	Y	--										
2.4.2	Smoke during cranking present or not (Specify white/ black smoke)								Target				Pass	Pass	Y	--										
2.4.3	Smoke level (visually) during start						%	New Development	Target				Pass	Pass	Y	--										
2.4.4	Smoke level (visually) during free acceleration at 120% after start						%		Target				Pass	Pass	Y	--										
2.4.5	Smoke level (visually) during free acceleration at 600% after start						%		Target				Pass	Pass	Y	--										
6.0	High altitude trials/ calibration								Heading						--											
6.2	Low idle governor calibration						rpm	New Development	Target				Pass	Pass	Y	--										
6.3.1	%Deterioration in Drivability with base calibration						%		Target				Pass	Pass	Y	--										
6.10.1	Turbocharger duty cycle mapping								Target				Pass	Pass	Y	--										
6.10.1.1	Smoke during driving present or not								Target				Pass	Pass	Y	--										
6.10.1.2	Assessment of turbocharger Whistling noise							New Development	Target				Pass	Pass	Y	--										
6.11.1	% Power Deterioration						%		Target				Pass	Pass	Y	--										
F	Onboard Diagnostics								Heading						--											
6.0	OBD Diagnostic - TML tool verification								Target				Pass	Pass	Y	--										
7.0	Deviate Strategies for Critical Faults								Target				Pass	Pass	Y	--										
8.0	Verification of OBD interface for Comprehensive Component Monitors								Target				Pass	Pass	Y	--										
Sign Off												Index for Status														
Responsibility												Name					Signature					Off target, no resolution				
Prepared by: PAT Engineer												Palash Das/Swapnil										Marginally off target, resolution planned/low risk				
Checked & Approved by: PAT Lead / Project Lead												Ajaykumar Jadhav / Sanjeev kumar										Met target, no concern				

PED evaluation of Signa 3523.T BSVI-II in comparison with Signa 3523.T BSVI-II Non-AC tested data as a part of AC regulation.

PED| 15.05.25

OUR CULTURE CREDO

AT TATA MOTORS

We are connecting aspirations by being bold in thought and action, owning every opportunity and challenge, Solving together as one team and engaging all our stakeholders with empathy.

We are **MORE WHEN ONE!**

BE BOLD

Taking calculated **risk** is key to making progress. We act with confidence and **agility** to accomplish our goals

OWN IT

Feeling and acting **empowered** is critical to drive results. We have an **Owner's Mind-set** and each of us takes full responsibility for the outcomes

SOLVE TOGETHER

Leveraging our collective genius while holding each other **accountable** helps us deliver the best. We **collaborate** proactively and transparently to achieve innovative solutions

BE EMPATHETIC

Embracing diversity makes us stronger for differences are opportunities to learn. We work with **passion to delight customers** and deliver greater success to our stakeholders

- ✓ Vehicle specification comparison
- ✓ Performance Data Summary
- ✓ Minimum and Maximum Gear Speed
- ✓ Wide open throttle performance
- ✓ Tractive effort
- ✓ Acceleration
- ✓ WOT drivability
- ✓ Bathtub drivability

Vehicle Specification Comparison

Sr. No.	Parameters	Base Vehicles BSVI-II	Test Vehicles BSVI-II
		Cowl 3523.T RDE BSVI	Signa 3523.T RDE BSVI
1	Engine	6 Cylinder 5.6 L / 230 HP BSVI	6 Cylinder 5.6 L / 230 HP BSVI
1.1	Engine Max. Power	220 HP	220 HP
1.2	Engine Max. Torque	925Nm @ 1000-1300rpm	925Nm @ 1000-1300rpm
1.3	Power to weight (HP/Ton)	6.28	6.28
1.4	Torque to weight (Nm/Ton)	26.42	26.42
1.5	Calibration ID	PF10040	PF10040
1.5a	CAL Used in Vehicle	XWTAA19D07	XWTAA19D07
1.6	Engine Calibration Maturity	100%	100%
1.7	Emission Cal Maturity	100%	100%
1.8	Regeneration Strategy	Auto	Auto
1.9	Urea Dosing	Yes	Yes
2.1	EOL Parameters	LBSC - 1900,2000,35000 kg, VAM - 3.0,2.5, 10, 40 Kmph, GDP - 50, 50 Kmph, RSG – 80 Kmph	LBSC - 1900,2000,35000 kg, VAM - 3.0,2.5, 10, 40 Kmph, GDP - 50, 50 Kmph, RSG – 80 Kmph
2	Fuel Economy Improvement Technologies		
2.1	Engine Cooling Fan	E-Viscous	E-Viscous
2.2	Fuel Economy Switch	4TC	4TC
2.3	Gear Shift Adviser	Yes	Yes
2.4	Accelerator Pedal Map	Yes	Yes
3	Other Parameters		
3.1	Fuel Tank type	Polymer	Polymer
3.2	Driveline	G 950 DD , FGR 9.36	G 950 DD , FGR 9.36
3.3	Gear Ratio	I-9.36, II-5.2, III-3.05, IV-1.84, V-1.33, VI-1,	I-9.36, II-5.2, III-3.05, IV-1.84, V-1.33, VI-1,
3.4	RAR	5.12	5.12
3.5	Tire size	Front – Rib 295/90R 20 Bridgestone Tyre Rear – Lug 295/90R20 Bridgestone Tyre	Front – Rib 295/90R 20 Bridgestone Tyre Rear – Lug 295/90R20 Bridgestone Tyre

Performance Data Summary

Parameter	Unit	Target wrt Base vehicle (AC ON)	Intended PALS	Simulated Data (Target)	Test Vehicle (Actual Data)		AC ON	Remarks
				Signa 3523.T OBDD II AC ON Target (Simulated)	Signa 3523.T BSVI-II_4TC With AC OFF RAR 5.12	Signa 3523.T BSVI-II_4TC With AC ON RAR 5.12	Test Vehicle % Status w.r.t Simulated AC ON Target (Simulated from Non-AC tested data)	
Acceleration								
0 - 60 kmph	sec	≤ 61.46	C	61.46	58.57	60.88	0.95%	
0 - 1000 m	sec	≤ 81.48	C	81.48	79.61	80.53	1.16%	
Driveability								
1st Gear(4-14)	sec	--	--	--	1.02	1.07	--	
2nd Gear(5-18)	sec			--	2.13	2.27		
3rd Gear(7-24)	sec			--	5.37	5.57		
4th Gear(10-30)	sec			--	13.83	14.56		
5th Gear(14-45)	sec			--	22.01	24.39		
6th Gear(20-60)	sec			--	26.07	30.25		
Rate of acceleration								
1st Gear(4-14)	kmph/sec	--	--	--	1.97	1.87	--	
2nd Gear(5-18)	kmph/sec			--	1.88	1.77		
3rd Gear(7-24)	kmph/sec			--	1.49	1.44		
4th Gear(10-30)	kmph/sec			--	1.01	0.96		
5th Gear(14-45)	kmph/sec			--	0.68	0.61		
6th Gear(20-60)	kmph/sec			--	0.54	0.46		
Bath-Tub Driveability								
4th Gear								
10-15 kmph	Sec	≤ 4.62	C	4.62	4.62	4.53	2.09%	
15-20 kmph	Sec	≤ 3.26	C	3.26	3.26	3.25	0.42%	
5th Gear								
20-25 kmph	Sec	≤ 6.35	C	6.35	5.80	6.15	3.17%	
25-30 kmph	Sec	≤ 4.04	C	4.04	3.84	4.02	0.68%	
30-35 kmph	Sec	≤ 4.21	C	4.21	3.96	4.11	2.39%	
35-40 kmph	Sec	--			4.51	4.69		
6th Gear								
25-30 kmph	Sec	≤ 10.44	C	10.44	9.81	10.18	2.45%	
30-35 kmph	Sec	≤ 6.78	C	6.78	6.99	6.74	0.59%	
35-40 kmph	Sec	≤ 5.85	C	5.85	5.44	5.47	6.45%	
7th Gear								
30-40 kmph	Sec	≤ 30.45	C	30.45	29.43	30.35	0.32%	
40-50 kmph	Sec	≤ 17.42	C	17.42	16.49	17.36	0.36%	
50-60 kmph	Sec	≤ 18.93	C	18.93	16.98	17.83	5.82%	
Gradability and Restartability								
Sustain Gradability	%	--	C		25.37		--	
Restart Gradability	%	--	C		18.80		--	

Signa 3523.T BSVI-II AC ON Performance. PALS position achieved "C" against "C"

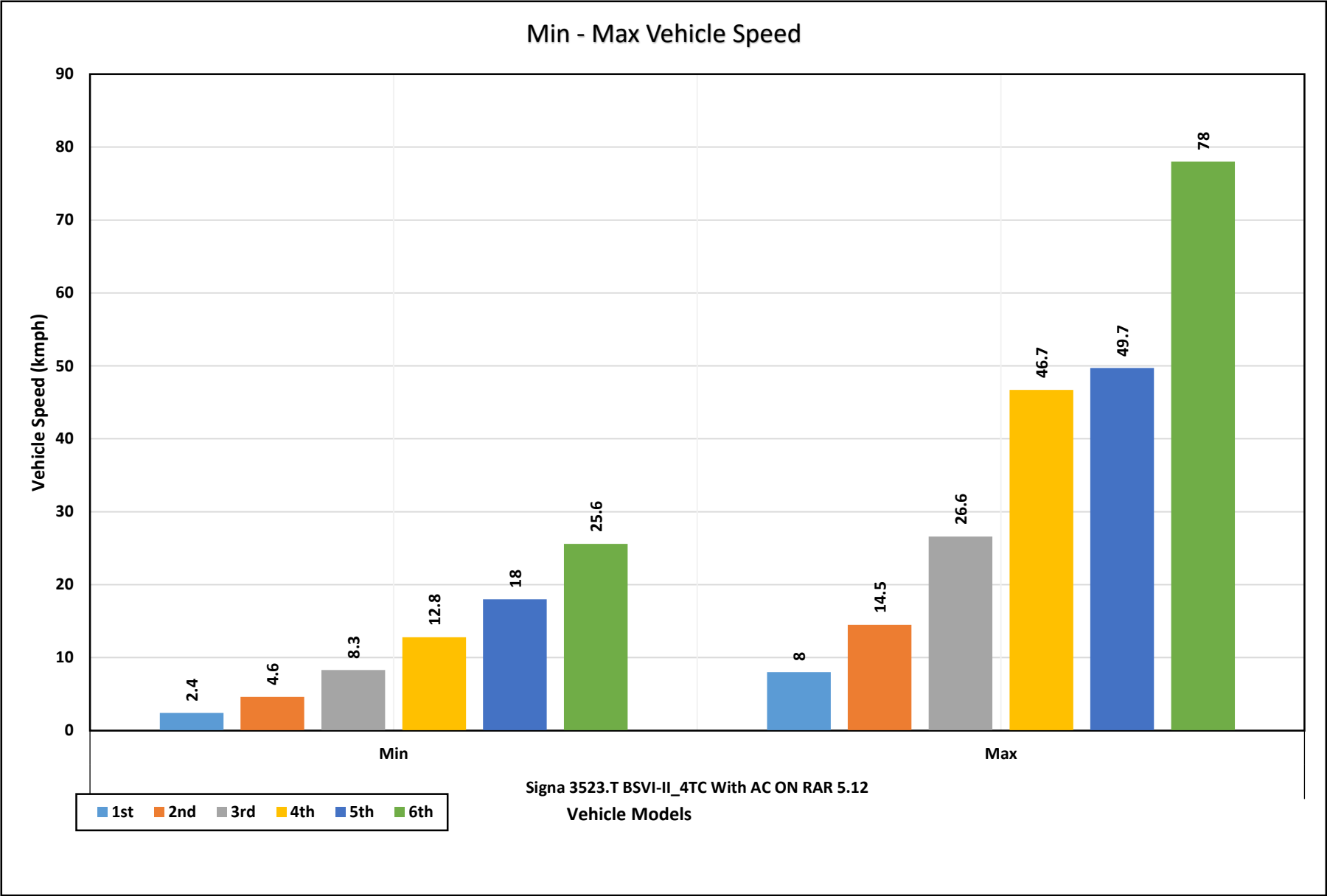
Acceleration

- Acceleration of Signa 3523.T BSVI-II was found to be better as compared to target in 0-60 kmph speed and 0-1000 meter band

Bath tub Drivability

- Drivability of Signa 3523.T BSVI-II was found to be better as compared to target in all gear.

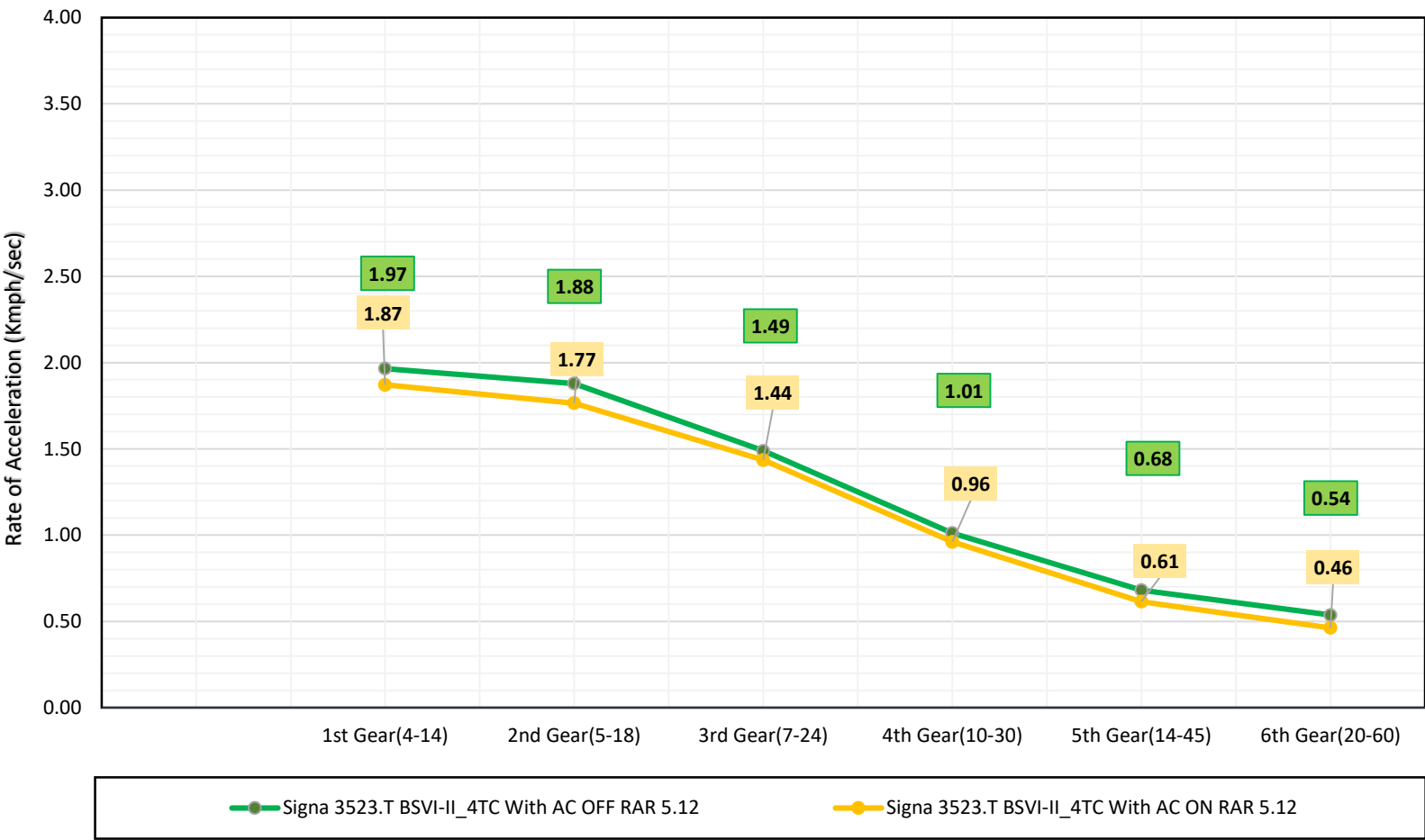
Minimum & Maximum Gear Wise speed Comparison



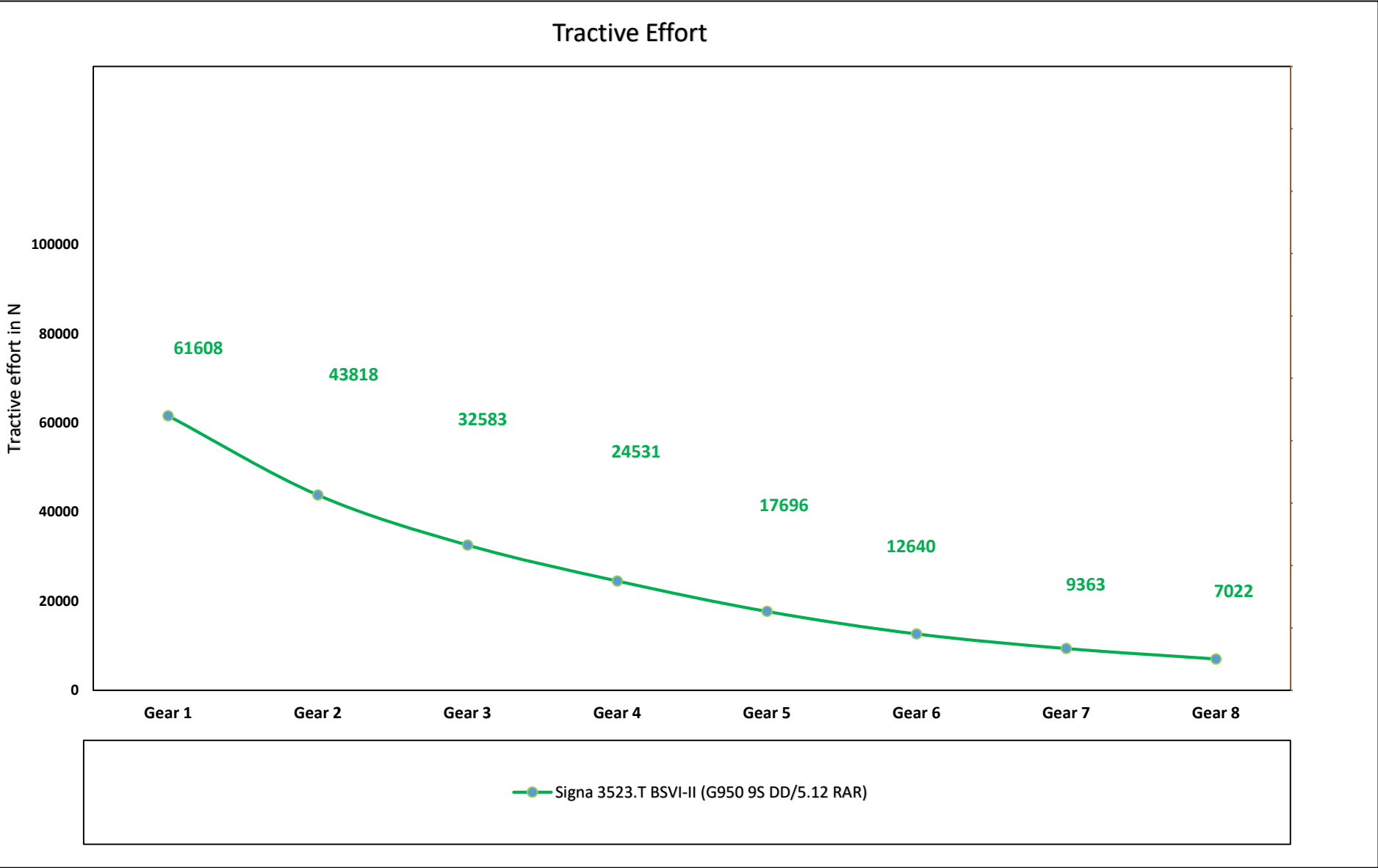
Gear	Signal 3523.T BSVI-II_4TC With AC ON RAR 5.12	
	Min	Max
1st	2.4	8
2nd	4.6	14.5
3rd	8.3	26.6
4th	12.8	46.7
5th	18	49.7
6th	25.6	78

Wide open throttle Performance

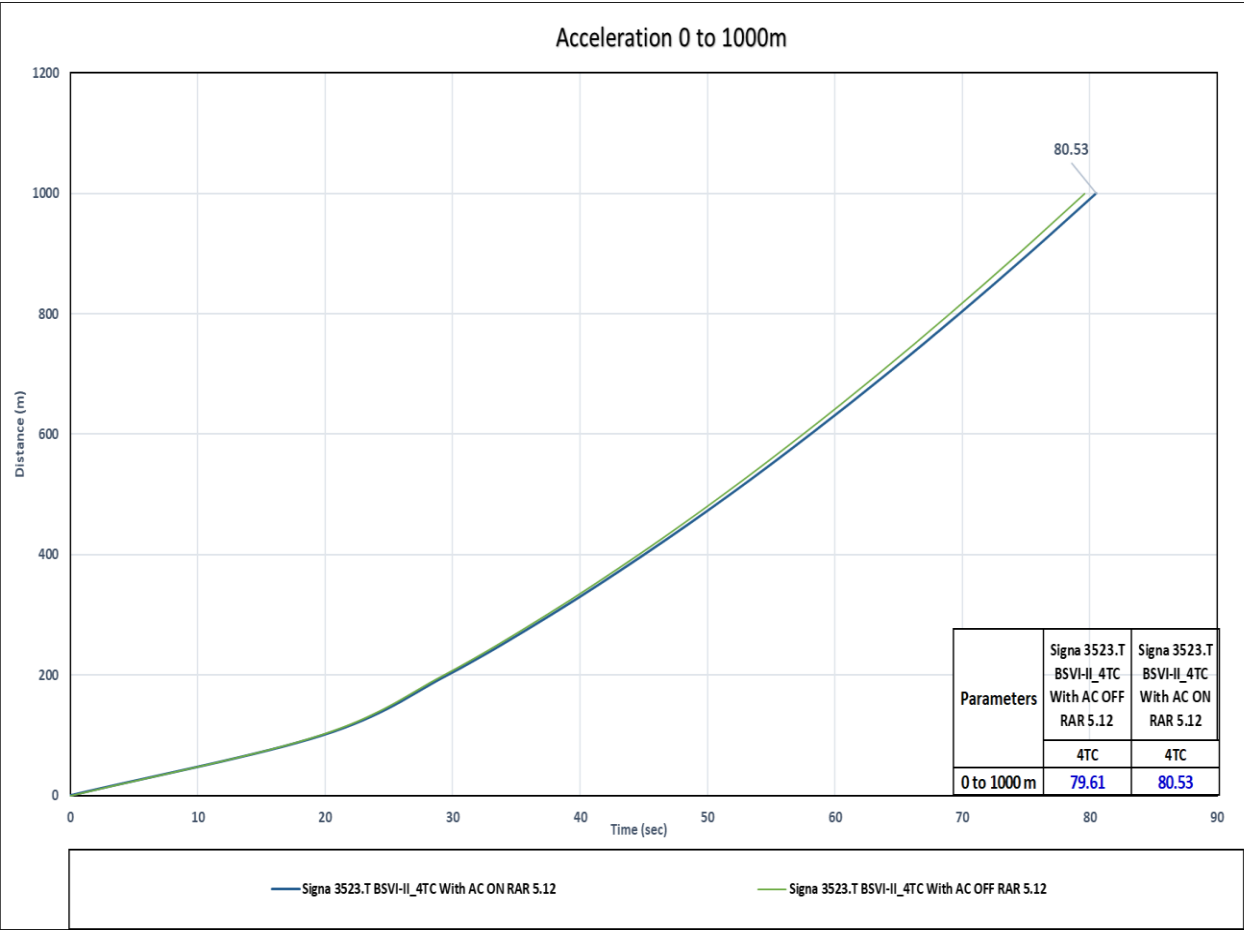
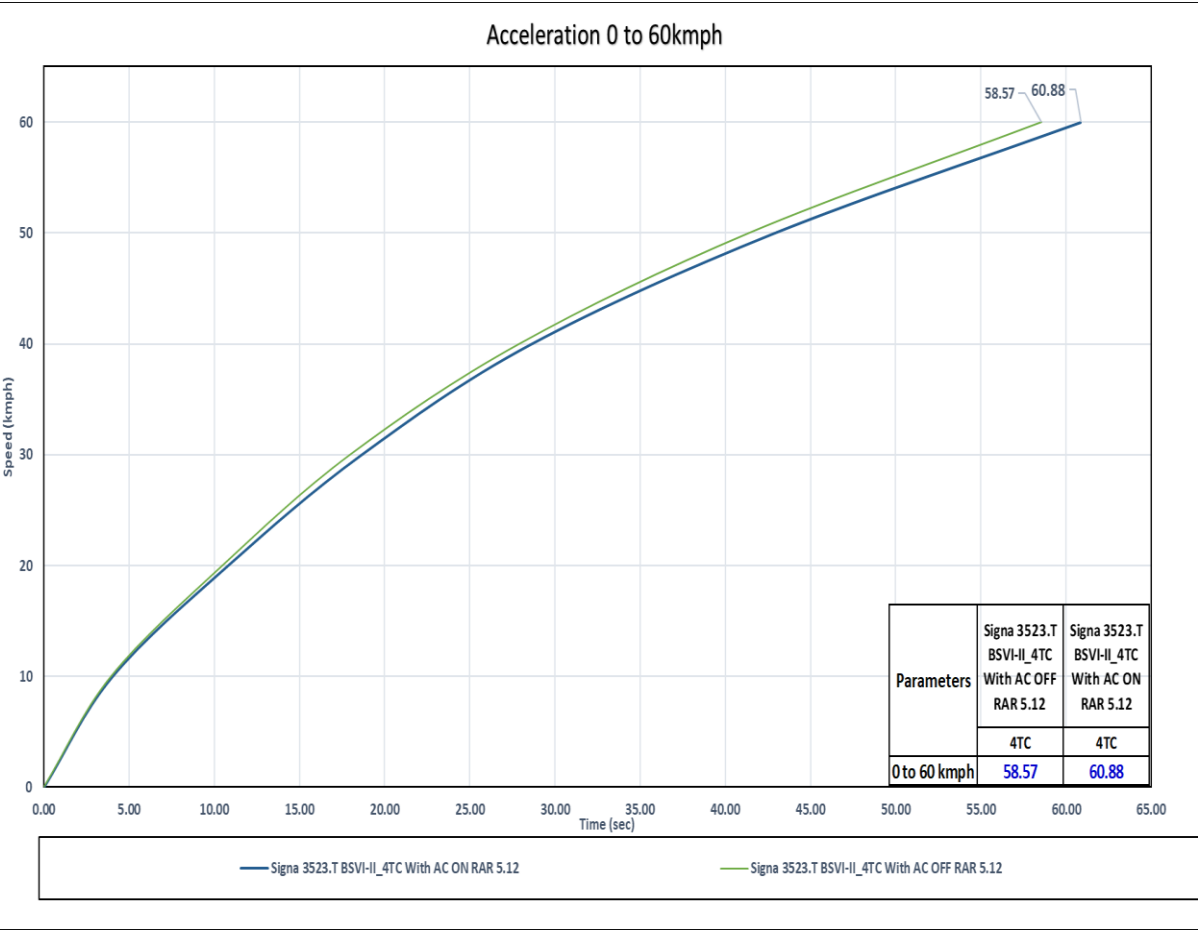
Gear Wise WOT Performance



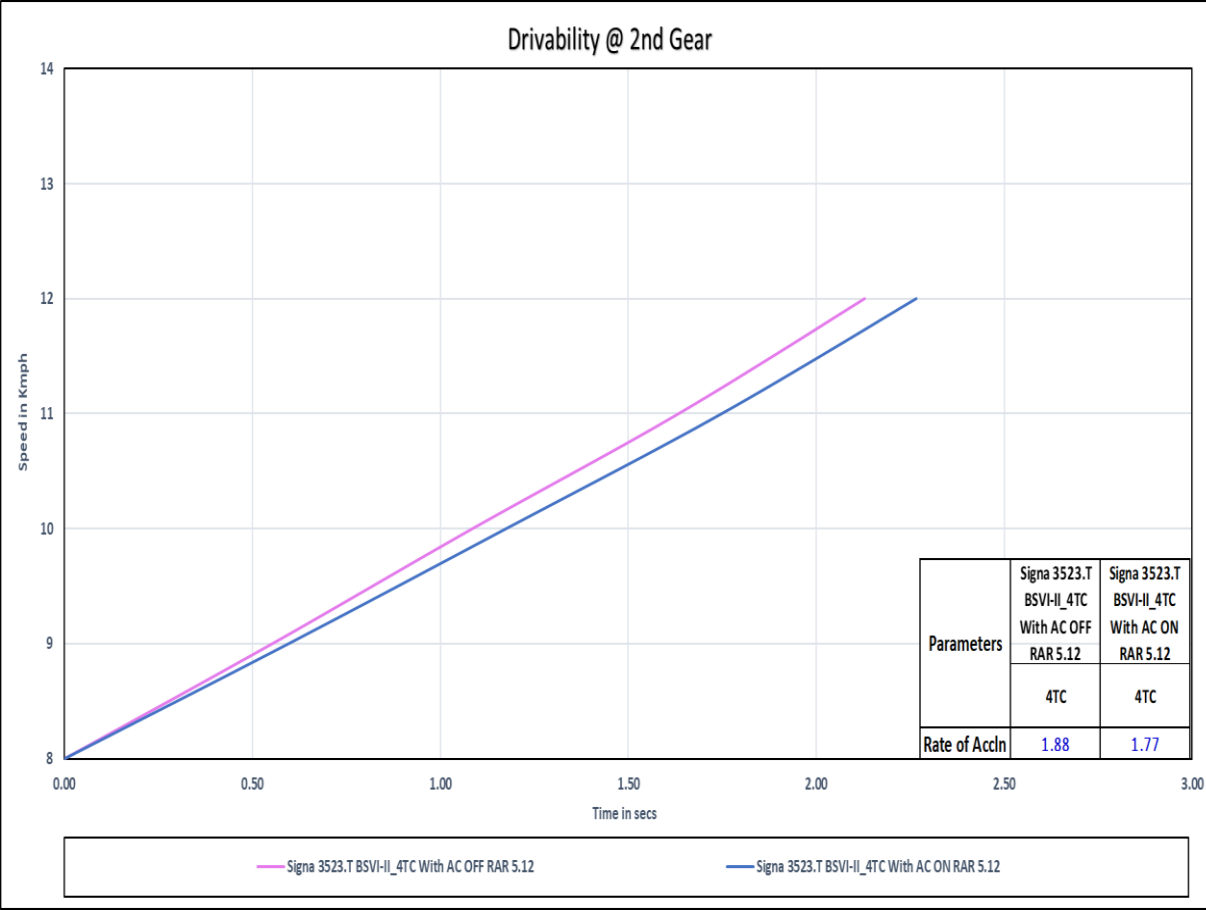
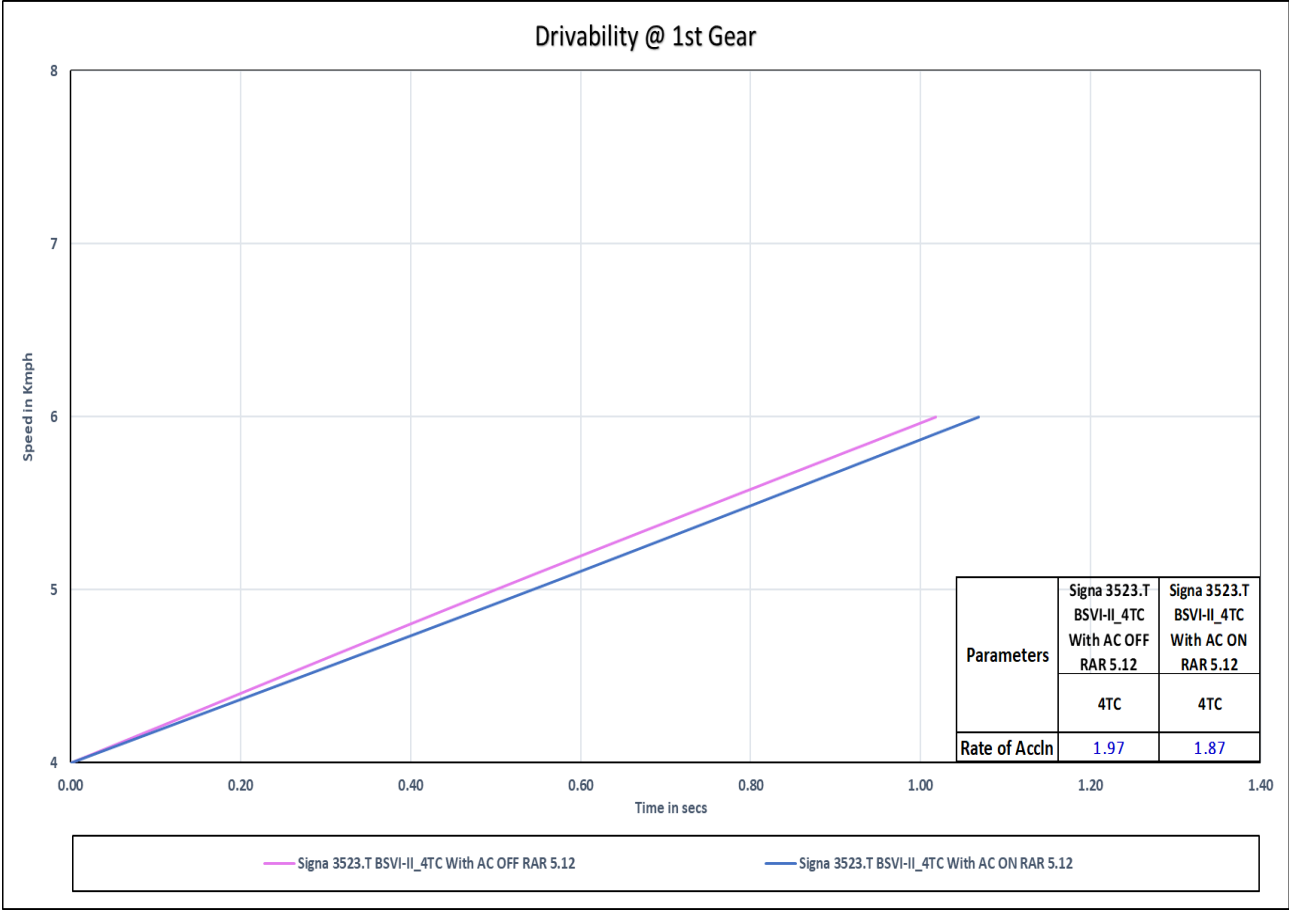
Gear	Signal 3523.T BSVI-II_4TC With AC OFF RAR 5.12	Signal 3523.T BSVI-II_4TC With AC ON RAR 5.12
1st Gear(4-14)	1.97	1.87
2nd Gear(5-18)	1.88	1.77
3rd Gear(7-24)	1.49	1.44
4th Gear(10-30)	1.01	0.96
5th Gear(14-45)	0.68	0.61
6th Gear(20-60)	0.54	0.46



Acceleration Performance – Through Gear

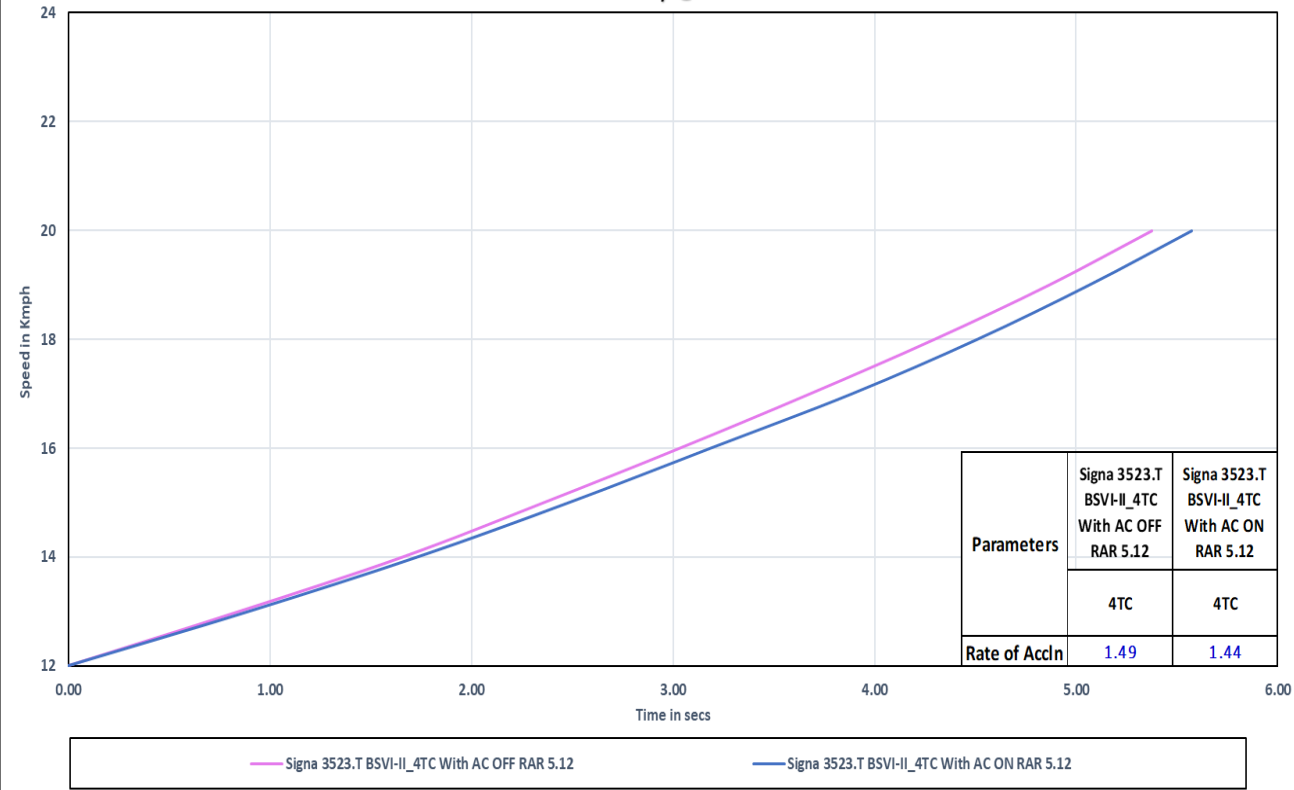


WOT Drivability @ 1st & 2nd Gear

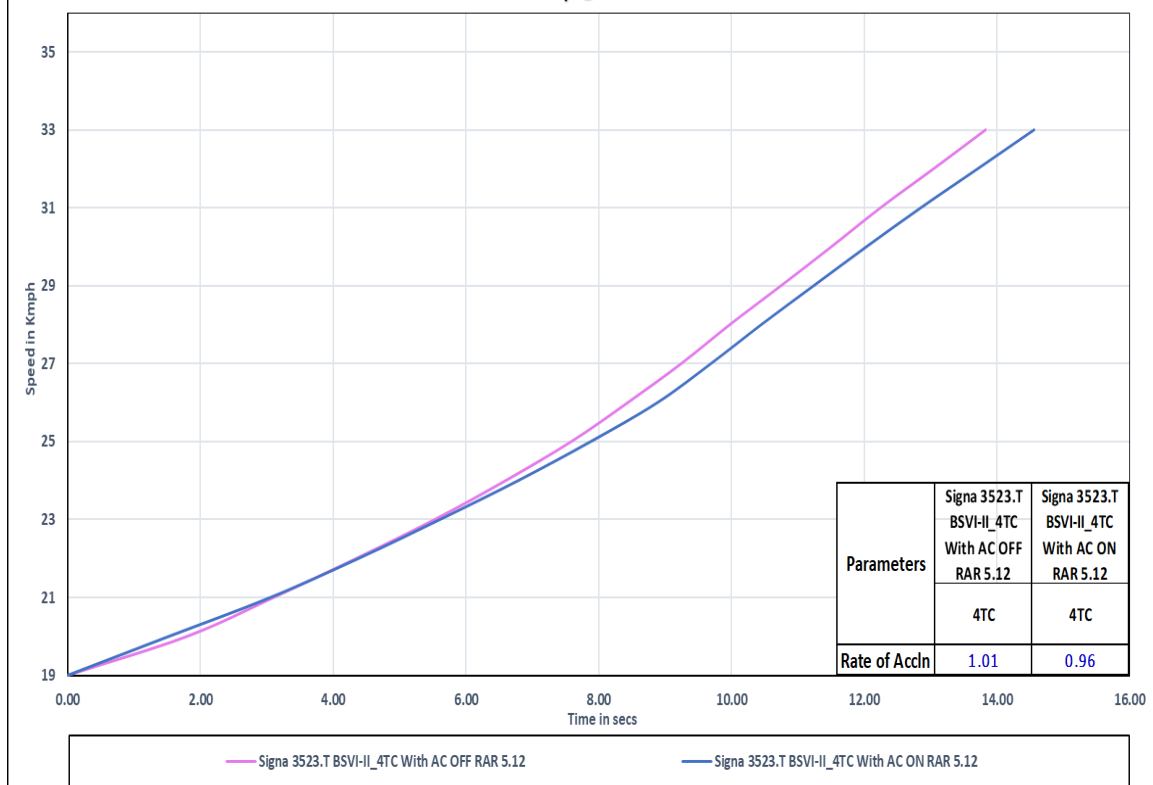


WOT Drivability @ 3rd & 4th Gear

Drivability @ 3rd Gear

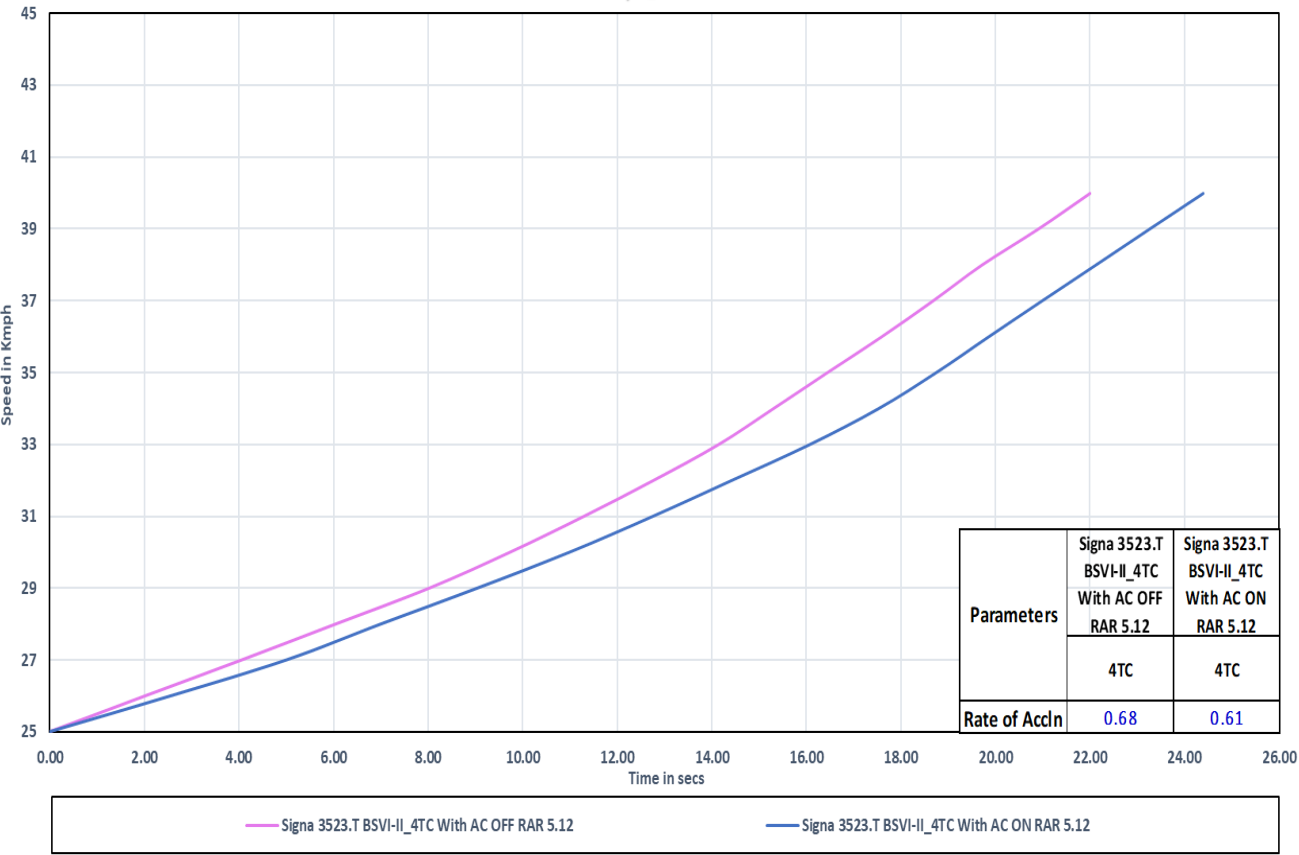


Drivability @ 4th Gear

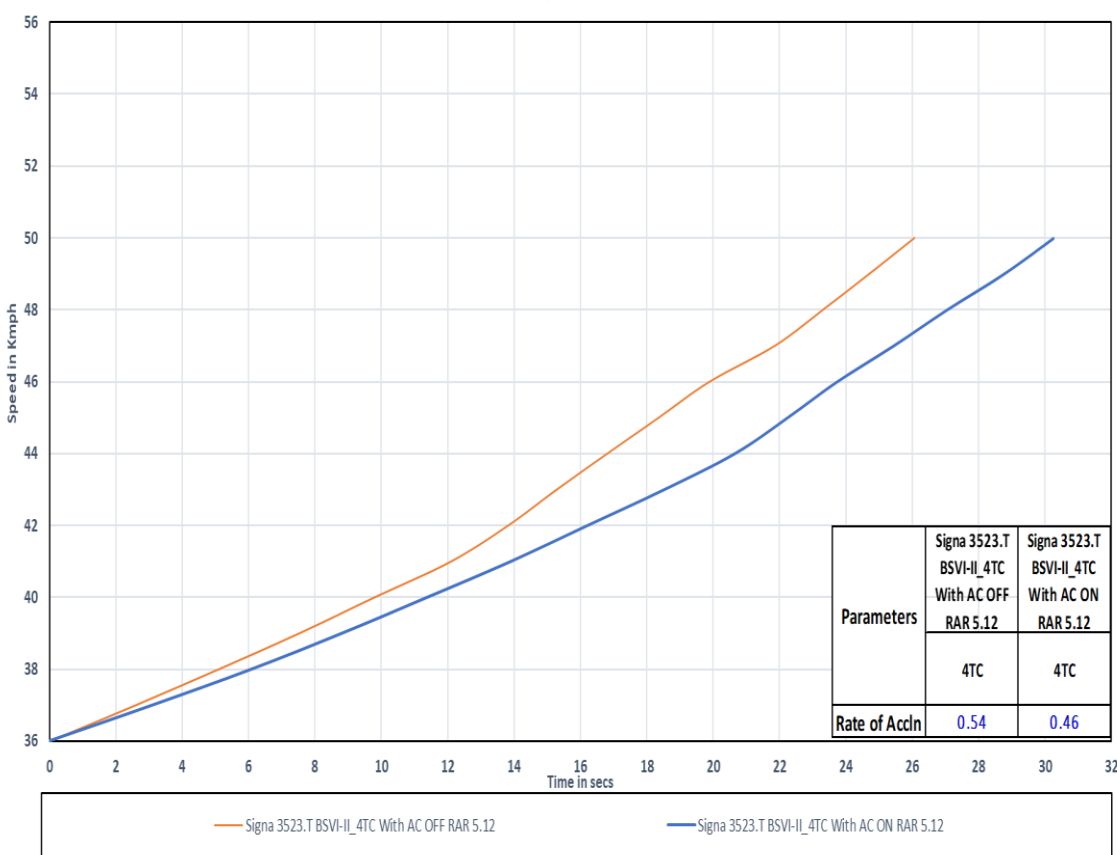


WOT Drivability @ 5th & 6th Gear

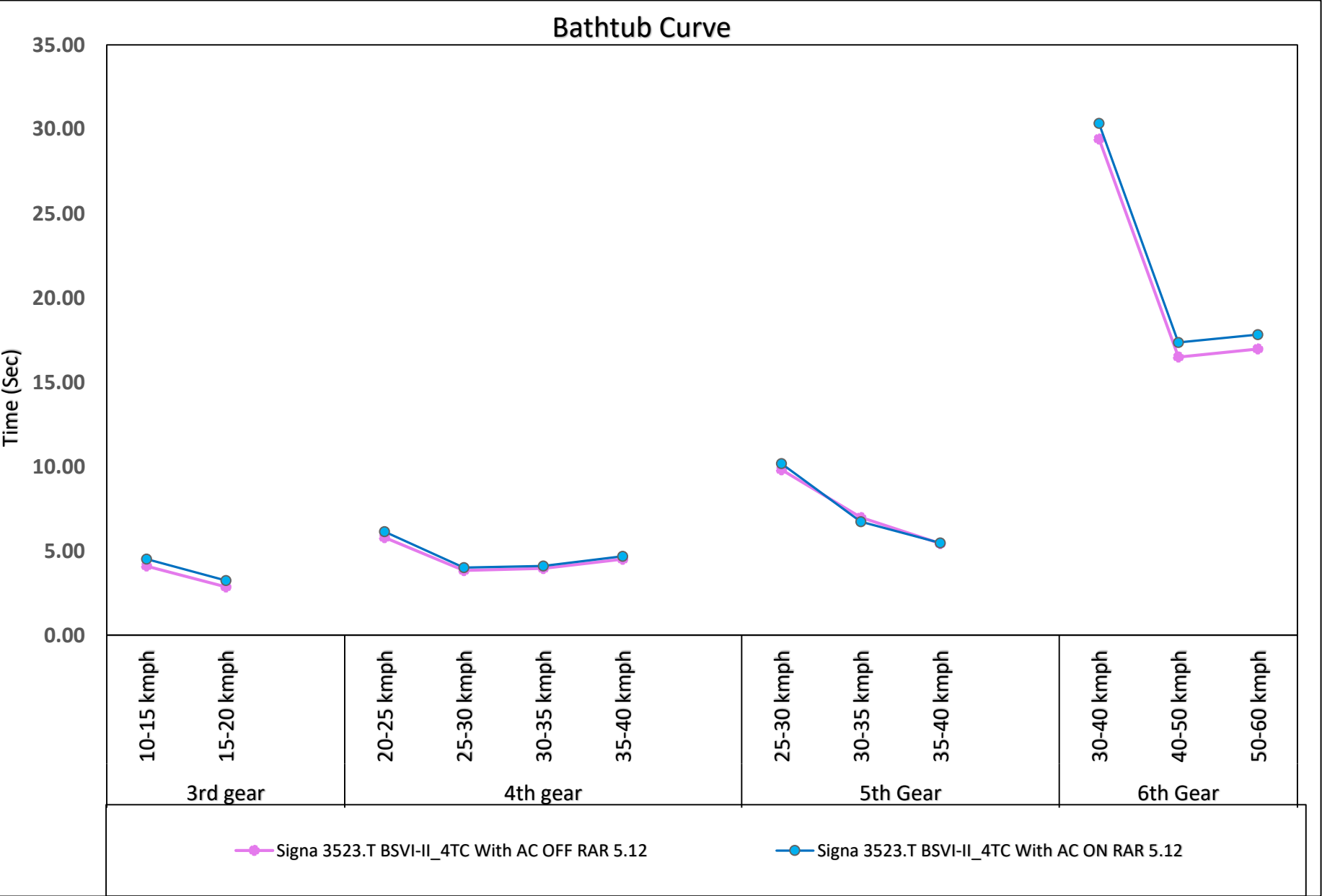
Drivability @ 5th Gear



Drivability @ 6th Gear



Bathtub Drivability Comparison with AC ON & OFF condition



Gear	Speed band	Signa 3523.T BSVI-II_4TC With AC OFF RAR 5.12	Signa 3523.T BSVI-II_4TC With AC ON RAR 5.12
		Time in Sec	Time in Sec
3rd gear	10-15 kmph	4.10	4.53
	15-20 kmph	2.87	3.25
4th gear	20-25 kmph	5.80	6.15
	25-30 kmph	3.84	4.02
	30-35 kmph	3.96	4.11
	35-40 kmph	4.51	4.69
5th Gear	25-30 kmph	9.81	10.18
	30-35 kmph	6.99	6.74
	35-40 kmph	5.44	5.47
6th Gear	30-40 kmph	29.43	30.35
	40-50 kmph	16.49	17.36
	50-60 kmph	16.98	17.83

Thank You!

TEST REQUISITION

Ref.	SIGNA 3523.T/PD/2025/02	Date:	27/04/2025
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To,

ERC Testing

Kindly arrange to take up for testing of the following component.

1	Part Description	TATA SIGNA 3523.T BSVI			
2	Part Number	52075256R,			
3	Drawing number	52075256R, 520700000001	Mod status	E;2,	Mod Date 15/03/2025
4	Vehicle Model / Project	Request from Mr.Bimal Kishor dated 05/01/2023			
5	WBS Number	P.24.EA.550AC REGULATION-TRUCK			
6	Supplier	NA			
7	Number of samples	-			
8	Inspection report	Enclosed	✓	If No commitment date:	
9	Material report	Enclosed	✓	If No commitment date:	
10	Suppliers test report	Enclosed	✓	If No commitment date:	
11	Reason for testing	First Source Development	✗	Weight Reduction	✗
		Alternate source development	✗	Cost Reduction	✗
		Data generation Test	✗	Regulatory test	✓
		Any other (pl. specify)	✗	New Development	✗
12	What testing required?				
13	Remarks	PD testing of SIGNA 3523.T BSVI with AC			
14	Peripheral Parts Availability	Yes			

For Testing Dept Use only	Test Item Number and Date(stamp)	Test Engineer

Authorized Signature:	
Name:	
Dept:	

Contact Person:	Kundan Kumar
Phone Number	

TATA MOTORS LIMITED ERC PUNE	TEST PLAN		DATE: 19/05/2025					
	VEHICLE TESTING DEPARTMENT							
Description: Performance evaluation of Signa 3523.T BSVI-II for AC regulation					Test Item No: UA001181			
					New	✓	Moderate	✗
					Major	✗	Minor	✗

Part No: 52075256R,		Modification No: E;2,	Enclosures: Annexure 1-5
Drawing No: 52075256R, 520700000001			
Make: NA		WBS Number: P.24.EA.550- AC REGULATION-TRUCK	Copies to :
Vehicle Type: Request from Mr.Bimal Kishor dated 05/01/2023		No. of Samples: -	
Test Requested by:	Testing-Outdoor-UA/01327		Equipments:
Test Requisition Reference:			
Reason for Test/s:	Regulatory Requirement		
Regulatory Requirement:	a) Component: TST/VT/WG/158		
	b) Vehicle: NA		

Sr. No.	Test description	Test specification	Acceptance criteria	No. of Samples	Test Responsibility	Proto Phase (Mule/ Alpha/ Beta/PP)	Remarks
1	Performance Evaluation of Signa 3523.T BSVI-II for AC regulation	TST/VT/WG/158	TST/VT/WG/158	1	Palash Das	Beta	completed

REMARKS: NA		INTERNAL REF:	NA
TEST TO FAILURE: No			
MEOST: Yes			
RELIABILITY TESTING: No			
BENCHMARKING TESTING: Yes			
DFMEA/FMEA DONE: Yes			
PREPARED BY		CHECKED BY	APPROVED BY
Palash Das			

PALASH DAS [TMCV, Engineering, Jamshedpur]

From: MANORANJAN SAHU [TMCV, Engineering, Jamshedpur]
Sent: 21 May 2025 17:32
To: SANJEEV KUMAR [TMCV, Engineering, Jamshedpur]; AJAYKUMAR JADHAV [TMCV, Engineering, Pune]; PANKAJ HAWARE [TMCV, Engineering, Pune]
Cc: PAVAN SINDGIKAR [TMCV, Engineering, Pune]; VISHNU DIXIT [TMCV, Engineering, Jamshedpur]; RAVINDRA DESHMUKH [TMCV, Engineering, Pune]; PRANAB SOU MANDAL [TMCV, Engineering, Jamshedpur]; DEBABRATA DAS [TMCV, Engineering, Jamshedpur]; SWAPNIL JADHAV [TMCV, Engineering, Pune]; PALASH DAS [TMCV, Engineering, Jamshedpur]; RAVI RANJAN [TMCV, Engineering, Jamshedpur]; SUJEET PRASAD [TTL, ER&D TML_iSource_CVBU]; SUBHANKAR BANERJEE [TTL, ER&D TML_iSource_CVBU]; RANA SINGH [TMCV, Engineering, Jamshedpur]; JITENDRA KUMAR SINGH [TMCV, Engineering, Jamshedpur]; AMIT KHOT [TTL, ER&D TML_iSource_CVBU]; SANTOSH KUMAR [TMCV, Engineering, Pune]; NAROTTAM PANKAJ [TMCV, Engineering, Pune]; KUNDAN KUMAR [TMCV, Engineering, Jamshedpur]
Subject: RE: HCV Truck_FE/PD Beta Validation FIR/Test Report- All lead & follower models name to be mention
Attachments: ODT PAT Extension Matrix.xlsx

Dear Sanjeev and PAT Leads,

Pls find the attached file with extension matrix and engine calibration details for Truck Models.

Regards
Manoranjan

From: MANORANJAN SAHU [TMCV, Engineering, Jamshedpur]
Sent: 16 May 2025 11:57
To: SANJEEV KUMAR [TMCV, Engineering, Jamshedpur] <sanjeev1.kumar@tatamotors.com>; AJAYKUMAR JADHAV [TMCV, Engineering, Pune] <ajaykumar_jadhav@tatamotors.com>; PANKAJ HAWARE [TMCV, Engineering, Pune] <pankaj.haware@tatamotors.com>
Cc: PAVAN SINDGIKAR [TMCV, Engineering, Pune] <pavan.s@tatamotors.com>; VISHNU DIXIT [TMCV, Engineering, Jamshedpur] <vishnud@tatamotors.com>; RAVINDRA DESHMUKH [TMCV, Engineering, Pune] <ravindra.deshmukh@tatamotors.com>; PRANAB SOU MANDAL [TMCV, Engineering, Jamshedpur] <pranab.soumandal@tatamotors.com>; DEBABRATA DAS [TMCV, Engineering, Jamshedpur] <DDD326082@tatamotors.com>; SWAPNIL JADHAV [TMCV, Engineering, Pune] <Swapnil.Jadhav@tatamotors.com>; PALASH DAS [TMCV, Engineering, Jamshedpur] <palash.das@tatamotors.com>; RAVI RANJAN [TMCV, Engineering, Jamshedpur] <ranjan.ravi@tatamotors.com>; SUJEET PRASAD [TTL, ER&D TML_iSource_CVBU] <sujeetp.ttl@tatamotors.com>; SUBHANKAR BANERJEE [TTL, ER&D TML_iSource_CVBU] <sb.ttl@tatamotors.com>; RANA SINGH [TMCV, Engineering, Jamshedpur] <Rana.Singh1@tatamotors.com>; JITENDRA KUMAR SINGH [TMCV, Engineering, Jamshedpur] <jitendrakumar.singh@tatamotors.com>; AMIT KHOT [TTL, ER&D TML_iSource_CVBU] <ak926960.ttl@tatamotors.com>; SANTOSH KUMAR [TMCV, Engineering, Pune] <santosh.kumar3@tatamotors.com>; NAROTTAM PANKAJ [TMCV, Engineering, Pune] <narottam.pankaj@tatamotors.com>
Subject: RE: HCV Truck_FE/PD Beta Validation FIR/Test Report- All lead & follower models name to be mention

Dear Sanjeev and PAT team leads,

This is with ref to the ESO team requirements for AC regulation and recent developments of 320 HP engines across the platform, request you to ensure **all lead & follower models name to be mentioned in PAT FIR/Test report accordingly** as per the attached variant matrix for Trucks.

Similar sheet will be shared by the other platform teams members.

CC: Rana, Jitendra , Santosh, and Narottam ji : Pls add your variant models on the attached matrix and share urgently.

Regards
Manoranjan

From: PRASHANT FATING [TMCV, Engineering, Pune] <prashant.fating@tatamotors.com>
Sent: 26 February 2025 13:36
To: MANORANJAN SAHU [TMCV, Engineering, Jamshedpur] <manoranjan.sahu@tatamotors.com>; NAROTTAM PANKAJ [TMCV, Engineering, Pune] <narottam.pankaj@tatamotors.com>; NIKHIL MATHUR [TMCV, Engineering, Pune] <NMM326121@tatamotors.com>; PARAG PHIRKE [TMCV, Engineering, Pune] <parag.phirke@tatamotors.com>; SUBHANKAR BANERJEE [TTL, ER&D TML_iSource_CVBU] <sb.ttl@tatamotors.com>; AMIT KHOT [TTL, ER&D TML_iSource_CVBU] <ak926960.ttl@tatamotors.com>; KUNDAN KUMAR [TMCV, Engineering, Jamshedpur] <kundan.kumar1@tatamotors.com>; PRITHVIRAJ RAY CHAUDHURI [TMCV, Engineering, Jamshedpur] <prithviraj.rc@tatamotors.com>; BIMAL KISHOR RAJAK [TTL, ER&D TML_iSource_CVBU] <bimalkis.ttl@tatamotors.com>; PRAMOD SAHU [TTL, ER&D TML_iSource_CVBU] <pramodsa.ttl@tatamotors.com>; PRAVIN KUMAR [TTL, ER&D TML_iSource_CVBU] <pravin.ttl@tatamotors.com>; DHIRAJ SINGH [TTL, ER&D TML_iSource_CVBU] <d0001001.ttl@tatamotors.com>
Cc: BHUSHAN KOTHARI [TMCV, Quality, Pune] <bkothari@tatamotors.com>; RAVINDRA DESHMUKH [TMCV, Engineering, Pune] <ravindra.deshmukh@tatamotors.com>; ASHISH DIGIKAR [TMCV, Quality, Pune] <ashish.digikar@tatamotors.com>; SANJEEV KUMAR [TMCV, Engineering, Jamshedpur] <sanjeev1.kumar@tatamotors.com>; SHASHWAT VISHWAKARMA [TTL, Engg Quality, ER&D TML_PO Onsite_EQ] <sv925274.ttl@tatamotors.com>; ROHIT SINKAR [TTL, Engg Quality, ER&D TML_PO Onsite_EQ] <rs926394.ttl@tatamotors.com>; SHUBHAM JAIN [TTL, ER&D TML_iSource_CVBU] <shubhamj.ttl@tatamotors.com>
Subject: HCV Truck_FE/PD Beta Validation FIR/Test Report- All lead & follower models name to be mention

Hello HCV Truck Team,

As per approved PAT extension matrix, **FE/PD testing (Beta Validation) to be done on -4830.T (6.7L), 3523.T (5.6) & 4221.T (5L),**

Request to you, Coordinate with ODT team for ...All lead & follower models name to be mention in PAT FIR/Test Report accordingly.

Reference list (6.7L, 5.6L - received from your side) attached -Lead/Follower model name

SIGNA AC PAT Extension Matrix (HCV- Truck, Tipper and Tractor)				Date: 25.09.2024 , REV - NR
	6.7 Ltr (Cummins) - Diesel	5.6 Ltr (Cummins) - Diesel	5Ltr (TML)- Diesel	Remarks
A1: Performance & Drivability	TBD: 5530.S / 4830.T / 4830.TK / 4225.T (G950) / SIGNA 2830.K REPTO	TBD: 4623.S / 3523.T / 4018.S	TBD: 5521.S / 4221.T / 2821.TK (RMC)	Based on the testing as per WCC, other tonnage node of 6.7 Ltr , 5.6 Ltr and 5 Ltr will get extended for all the platform
A2: Fuel Economy	TBD: 5530.S / 4830.T / 4830.TK/ SIGNA 2830.K REPTO	TBD: 4623.S / 3523.T / 2818.TK (RMC)	TBD: 5521.S / 4221.T / 2821.TK (RMC)	Based on the testing as per WCC, other tonnage node of 6.7 Ltr , 5.6 Ltr and 5 Ltr will get extended for all the platform

COWL AC PAT Extension Matrix (Truck)				Date: 25.09.2024 , REV - NR
	6.7 Ltr (Cummins) - Diesel	5.6 Ltr (Cummins) - Diesel	5Ltr (TML)- Diesel	Remarks
A1: Performance & Drivability	C/O From Base SIGNA	C/O From Base SIGNA	C/O From Base SIGNA	Based on the testing as per WCC, other tonnage node of 6.7 Ltr , 5.6 Ltr and 5 Ltr will get extended for all the platform
A2: Fuel Economy	C/O From Base SIGNA	C/O From Base SIGNA	C/O From Base SIGNA	Based on the testing as per WCC, other tonnage node of 6.7 Ltr , 5.6 Ltr and 5 Ltr will get extended for all the platform

Regards,
Prashant Fating
ESO-Engineering Signoff

ODT FIR Extension Matrix					
Sl No	Vehicle PAT	Engine	Lead vehicle / Vehicle Tested	Truck Models	Engine Cal Id
1	FE	6.7L (300HP)	SIGNA/LPT 4830.T	SIGNA/LPT 4830.T	PE10190
2				SIGNA/LPT 4930.T	PE10190
3				SIGNA/LPT 4832.T	PE10210
4				SIGNA/LPT 4932.T	PE10210
5				SIGNA/LPT 4425.T	PE10192
6				SIGNA/LPT 4225.T	PE10192
7				SIGNA/LPT 3725.T	PE10192
8				SIGNA/LPT 3525.T	PE10192
9				SIGNA/LPT 3125.T	PE10192
10		5.6L (230HP)	SIGNA/LPT 3523.T	SIGNA/LPT 3723.T	PF10040
11				SIGNA/LPT 3523.T	PF10040
12				SIGNA/LPT 3123.T	PF10040
13				SIGNA/LPT 3023.T	PF10040
14				SIGNA/LPT 2823.T	PF10040/PF10046
15				SIGNA/LPT 2623.T	PF10040
16		5L (210HP)	SIGNA/LPT 3521.T	SIGNA/LPT 3721.T	
17				SIGNA/LPT 3521.T	
18				SIGNA/LPT 3121.T	
19				SIGNA/LPT 3021.T	
20				SIGNA/LPT 2821.T	
21				SIGNA/LPT 2621.T	

Annexure 6

GURPRIT SINGH [TMCV, Engineering, Pune]

From: GURPRIT SINGH [TMCV, Engineering, Pune]
Sent: 21 May 2025 17:16
To: RANA SINGH [TMCV, Engineering, Jamshedpur]; AJAYKUMAR JADHAV [TMCV, Engineering, Pune]
Cc: SWAPNIL JADHAV [TMCV, Engineering, Pune]; SANJEEV KUMAR [TMCV, Engineering, Jamshedpur]; Bokam Surya Sashikanth; Sweta Kumari; Ashwani Kumar Gupta; Raj Mohanty; Anil S Patil; Anish Kumar; DHEERAJ KUMAR [TMCV, Engineering, Jamshedpur]
Subject: RE: Calibration no for tipper extension matrix

Dear Rana Ji
As per attached excel, we are considering below Cal ids against extension matrix.

ODT FIR Extension Matrix					
				Tipper Models	
SI No	Vehicle PAT	Engine	Lead vehicle/Vehicle Tested	Variant	CAL ID
1	FE	6.7L (300HP)	SIGNA 4830.TK	SIGNA 4830.TK	PE10197.01
2				SIGNA 4230.TK	PE10197.01
3				SIGNA 3530.K/TK	PE10185.00
4				SIGNA 2830.TK/TK	PE10185.00
5				SIGNA 4832.TK	PE10212.00
6				SIGNA 4232.TK	PE10212.00
7				SIGNA 3532.TK/TK	PE10211.01
8				SIGNA 2832.K/TK	PE10211.01
9				SIGNA 2825.K/TK	PE10188.00
10		6.7L (300HP)	SIGNA 2830.TK REPTO	SIGNA 2830.TK REPTO	PE10185.00
				SIGNA 2832.K REPTO	PE10211.01
11		5.6L (230HP)	SIGNA/LPT 3523.T	SIGNA 2823.K/TK	PE10050.00
13		5.6L (180HP)	SIGNA 2818.K	SIGNA 2818.K	PE10051.00
14		5L (210HP)	SIGNA 2821.K	LPK2821.K	
15		5.7L CNG	Signa 2820.K	Signa 2820.K	

ODT FIR Extension Matrix					
				Tipper Models	
SI No	Vehicle PAT	Engine	Lead vehicle/Vehicle Tested	Variant	CAL ID
1	PD	6.7L (300HP)	SIGNA 4830.TK	SIGNA 4830.TK	PE10197.01
2				SIGNA 4230.TK	PE10197.01
3				SIGNA 3530.K/TK	PE10185.00
4				SIGNA 2830.K/TK	PE10185.00
				SIGNA 2830.K REPTO	PE10185.00
5				SIGNA 4832.TK	PE10212.00
6				SIGNA 4232.TK	PE10212.00
7				SIGNA 3532.K/TK	PE10211.01
8				SIGNA 2832.K/TK	PE10211.01
9				SIGNA 2832.K REPTO	PE10211.01
10		6.7L (25HP)	SIGNA/LPT 4225.T	SIGNA 2825.K/TK	PE10188.00
11		5.6L (230HP)	SIGNA/LPT 3523.T	SIGNA 2823.K/TK	PE10050.00
13		5.6L (180HP)	SIGNA 4018.S	SIGNA 2818.K	PE10051.00
14		5L (210HP)	SIGNA 2821.K	LPK2821.K	
15		5.7L CNG	Signa 2820.T	Signa 2820.K	

Regards
Gurprit Singh

From: RANA SINGH [TMCV, Engineering, Jamshedpur] <Rana.Singh1@tatamotors.com>
Sent: 21 May 2025 17:05
To: AJAYKUMAR JADHAV [TMCV, Engineering, Pune] <ajaykumar_jadhav@tatamotors.com>; GURPRIT SINGH [TMCV, Engineering, Pune] <Gurprit.Singh@tatamotors.com>
Cc: SWAPNIL JADHAV [TMCV, Engineering, Pune] <Swapnil.Jadhav@tatamotors.com>; SANJEEV KUMAR [TMCV, Engineering, Jamshedpur] <sanjeev1.kumar@tatamotors.com>; Bokam Surya Sashikanth <bokam.surya.sashikanth@cummins.com>; Sweta Kumari <sweta.kumari@cummins.com>; Ashwani Kumar Gupta <ashwani.kumar.gupta@cummins.com>; Raj Mohanty <raj.mohanty@cummins.com>; Anil S Patil <anil.s.patil@cummins.com>; Anish Kumar <kumar.anish@cummins.com>; DHEERAJ KUMAR [TMCV, Engineering, Jamshedpur] <kumar.dheeraj@tatamotors.com>
Subject: RE: Calibration no for tipper extension matrix
Importance: High

Hi Ajay/Gurprit

Refer attached matrix for with CAL ID.

With Regards

From: AJAYKUMAR JADHAV [TMCV, Engineering, Pune] <ajaykumar_jadhav@tatamotors.com>
Sent: 21 May 2025 16:08
To: RANA SINGH [TMCV, Engineering, Jamshedpur] <Rana.Singh1@tatamotors.com>
Cc: SWAPNIL JADHAV [TMCV, Engineering, Pune] <Swapnil.Jadhav@tatamotors.com>; SANJEEV KUMAR [TMCV, Engineering, Jamshedpur] <sanjeev1.kumar@tatamotors.com>; Bokam Surya Sashikanth <bokam.surya.sashikanth@cummins.com>; Sweta Kumari <sweta.kumari@cummins.com>; Ashwani Kumar Gupta <ashwani.kumar.gupta@cummins.com>; Raj Mohanty <raj.mohanty@cummins.com>; Anil S Patil <anil.s.patil@cummins.com>; Anish Kumar <kumar.anish@cummins.com>; GURPRIT SINGH [TMCV, Engineering, Pune] <Gurprit.Singh@tatamotors.com>
Subject: RE: Calibration no for tipper extension matrix

Dear Rana,
Kindly check and verify as we are going to updated report based on your feedback.

Regards,
Ajay

From: Anish Kumar <kumar.anish@cummins.com>
Sent: 21 May 2025 16:07
To: GURPRIT SINGH [TMCV, Engineering, Pune] <Gurprit.Singh@tatamotors.com>; RANA SINGH [TMCV, Engineering, Jamshedpur] <Rana.Singh1@tatamotors.com>
Cc: AJAYKUMAR JADHAV [TMCV, Engineering, Pune] <ajaykumar_jadhav@tatamotors.com>; SWAPNIL JADHAV [TMCV, Engineering, Pune] <Swapnil.Jadhav@tatamotors.com>; SANJEEV KUMAR [TMCV, Engineering, Jamshedpur] <sanjeev1.kumar@tatamotors.com>; Bokam Surya Sashikanth <bokam.surya.sashikanth@cummins.com>; Sweta Kumari <sweta.kumari@cummins.com>; Ashwani Kumar Gupta <ashwani.kumar.gupta@cummins.com>; Raj Mohanty <raj.mohanty@cummins.com>; Anil S Patil <anil.s.patil@cummins.com>
Subject: RE: Calibration no for tipper extension matrix

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Hello Team,

Please refer below calibration IDs for vehicles models, Kindly note that we have added one column for engine configuration based on Rana’s input and added calibration code against that.

Calibration code changes if the engine configuration changes.

ODT FIR Extension Matrix						
				Tipper Models		
SI No	Vehicle PAT	Engine	Lead vehicle/Vehicle Tested	Variant		Calibration ID
1	FE	6.7L (300HP)	SIGNA 4830.TK	SIGNA 4830.TK	VI+Front EMP+3”	PE10197
2				SIGNA 4230.TK	VI+front EMP+3”	PE10197
3				SIGNA 3530.K/TK	Coal+Front EMP+4”	PE10185
4				SIGNA 2830.K/TK	Coal+Front EMP+4”	PE10185
5				SIGNA 4832.TK	VI+Front EMP+3”	PE10213
6				SIGNA 4232.TK	VI+front EMP+3”	PE10213
7				SIGNA 3532.K/TK	Coal+Front EMP + 4”	PE10234
8				SIGNA 2832.K/TK	Coal+Front EMP + 4”	PE10234
9				SIGNA 2825.K/TK	VI + Tri-EGTS+front EMP	PE10188
10		6.7L (300HP)	SIGNA 2830.TK REPTO	SIGNA 2832.K REPTO		PE10185 (same as normal tipper)
11		5.6L (230HP)	SIGNA/LPT 3523.T	SIGNA 2823.K/TK	Coalescer breather+Tri EGTS+Rear Emp	PF10050
13		5.6L (180HP)	SIGNA 2818.K	SIGNA 2818.K	Coal+Tri-EGTS	PF10051 (RMC)
14		5L (210HP)	SIGNA 2821.K	LPK2821.K	TML	NA
15		5.7L CNG	Signa 2820.K	Signa 2820.K	TML	NA

Best Regards
Anish Kumar
Technical Specialist – FE Cell
Cell: 9264433836

From: Gurprit.Singh@tatamotors.com <Gurprit.Singh@tatamotors.com>
Sent: 21 May 2025 15:10
To: Rana.Singh1@tatamotors.com
Cc: ajaykumar_jadhav@tatamotors.com; Swapnil.Jadhav@tatamotors.com; sanjeev1.kumar@tatamotors.com; Anish Kumar <kumar.anish@cummins.com>; Bokam Surya Sashikanth <bokam.surya.sashikanth@cummins.com>; Sweta Kumari <sweta.kumari@cummins.com>
Subject: Calibration no for tipper extension matrix

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Dear Ranaji

We discussed, please provide calibration no for Tipper extension matrix.

ODT FIR Extension Matrix					
				Tipper Models	
SI No	Vehicle PAT	Engine	Lead vehicle/Vehicle Tested	Variant	Calibration ID
1	FE	6.7L (300HP)	SIGNA 4830.TK	SIGNA 4830.TK	
2				SIGNA 4230.TK	
3				SIGNA 3530.K/TK	
4				SIGNA 2830.K/TK	
5				SIGNA 4832.TK	
6				SIGNA 4232.TK	
7				SIGNA 3532.K/TK	
8				SIGNA 2832.K/TK	
9				SIGNA 2825.K/TK	
10		6.7L (300HP)	SIGNA 2830.TK REPTO	SIGNA 2832.K REPTO	
11		5.6L (230HP)	SIGNA/LPT 3523.T	SIGNA 2823.K/TK	
13		5.6L (180HP)	SIGNA 2818.K	SIGNA 2818.K	
14		5L (210HP)	SIGNA 2821.K	LPK2821.K	
15		5.7L CNG	Signa 2820.K	Signa 2820.K	

ODT FIR Extension Matrix					
				Tipper Models	
SI No	Vehicle PAT	Engine	Lead vehicle/Vehicle Tested	Variant	Calibration ID
1	PD	6.7L (300HP)	SIGNA 4830.TK	SIGNA 4830.TK	
2				SIGNA 4230.TK	
3				SIGNA 3530.K/TK	
4				SIGNA 2830.K/TK	
5				SIGNA 2830.K REPTO	
6				SIGNA 4832.TK	
7				SIGNA 4232.TK	
8				SIGNA 3532.K/TK	
9				SIGNA 2832.K/TK	
10				SIGNA 2832.K REPTO	
11		6.7L (25HP)	SIGNA/LPT 4225.T	SIGNA 2825.K/TK	
12		5.6L (230HP)	SIGNA/LPT 3523.T	SIGNA 2823.K/TK	

13		5.6L (180HP)	SIGNA 4018.S	SIGNA 2818.K	
14		5L (210HP)	SIGNA 2821.K	LPK2821.K	
15		5.7L CNG	Signa 2820.T	Signa 2820.K	

Regards
Gurprit Singh

From: GURPRIT SINGH [TMCV, Engineering, Pune]
Sent: 21 May 2025 12:02
To: Anish Kumar <kumar.anish@cummins.com>; Sweta Kumari <sweta.kumari@cummins.com>; Bokam Surya Sashikanth <bokam.surya.sashikanth@cummins.com>
Cc: AJAYKUMAR JADHAV [TMCV, Engineering, Pune] <ajaykumar_jadhav@tatamotors.com>; SWAPNIL JADHAV [TMCV, Engineering, Pune] <Swapnil.Jadhav@tatamotors.com>; SANJEEV KUMAR [TMCV, Engineering, Jamshedpur] <sanjeev1.kumar@tatamotors.com>
Subject: RE: HCV_AC project PED validation status

Dear Anish
Request you to also provide the details of the following models

Model	ENG	ERC No	Calibration ID	CAL Used in Vehicle
SIGNA 4230.TK				
SIGNA 3530.K/TK				
SIGNA 2830.K/TK				
SIGNA 4832.TK				
SIGNA 4232.TK				
SIGNA 3532.K/TK				
SIGNA 2832.K/TK				
SIGNA 2825.K/TK				
Signa 2823 RMC				

Regards
Gurprit Singh

From: Anish Kumar <kumar.anish@cummins.com>
Sent: 20 May 2025 15:22
To: SWAPNIL JADHAV [TMCV, Engineering, Pune] <Swapnil.Jadhav@tatamotors.com>; Sweta Kumari <sweta.kumari@cummins.com>; Bokam Surya Sashikanth <bokam.surya.sashikanth@cummins.com>; AJAYKUMAR JADHAV [TMCV, Engineering, Pune] <ajaykumar_jadhav@tatamotors.com>; HCV PED <MHCVPED@tatamotors.onmicrosoft.com>; GURPRIT SINGH [TMCV, Engineering, Pune] <Gurprit.Singh@tatamotors.com>
Subject: RE: HCV_AC project PED validation status

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Hello Ajay/Swapnil,

Please refer below CAL-ID for priority tested vehicles.

Sr. No.	Model	ENG	ERC Number	Calibration ID	CAL Used in Vehicle
1	Signa 5530.S 6x4	6.7L 300HP	ERC1283	PE10143.03	XYPAA17D04
2	Cowl 4830.T		EJ00913	PE10182.01	PE10182.01
3	Signa 4830.T		ERC1210	PE10190.02	XYRAA17E85
4	Signa 4830.TK		ERC1261	PE10003.15	XYPAA17D91

5	Signa 2830.K REPTO		ERC1237	PE10206.02	XYPAA17E07
6	Signa 4225.T (G950)	250HP	ERC1282	XYPAA17D17	XYPAA17F01
7	Signa 3523.T	5.6L 230 HP	ERC1309	PF10040.02	XWTAA19D07
8	Signa 4623.S		EJ00935	PF10002.09	XWTAA19D14
9	Signa 4023.S - 750NM		EJ00931	PF10070	XWTAA19D10
10	Signa 2818.TK RMC	5.6L 180HP - 700Nm	EJ00936	PF10019.02	XWTAA13D15

Best Regards
Anish Kumar
Technical Specialist – FE Cell
Cell: 9264433836

From: Swapnil.Jadhav@tatamotors.com <Swapnil.Jadhav@tatamotors.com>
Sent: 20 May 2025 12:30
To: Anish Kumar <kumar.anish@cummins.com>; Sweta Kumari <sweta.kumari@cummins.com>; Bokam Surya SashiKanth <bokam.surya.sashikanth@cummins.com>
Cc: ajaykumar_jadhav@tatamotors.com; MHCVPED@tatamotors.onmicrosoft.com
Subject: RE: HCV_AC project PED validation status

EXTERNAL SENDER: This email originated outside of Cummins. Do not click links or open attachments unless you verify the sender and know the content is safe.

Dear Surya,

Instead of revision number please mention CVN number

Best Regards,
Swapnil Jadhav
PED - HCV
ERC Pune - Vehicle Engg. (CVBU)
Mo. No. - +91 9922001494



From: SWAPNIL JADHAV [TMCV, Engineering, Pune]
Sent: 20 May 2025 11:26
To: Anish Kumar <kumar.anish@cummins.com>; Sweta Kumari <sweta.kumari@cummins.com>
Cc: AJAYKUMAR JADHAV [TMCV, Engineering, Pune] <ajaykumar_jadhav@tatamotors.com>; HCV PED <MHCVPED@tatamotors.onmicrosoft.com>
Subject: FW: HCV_AC project PED validation status

Dear Anish,

As discussed, PI revert below table with Calibration ID & revision number.

Its Urgent.

Sr. No.	Model	ENG	Calibration ID	CVN
---------	-------	-----	----------------	-----

1	Signa 5530.S 4X2	6.7L 300HP		
2	Cowl 4830.T			
3	Signa 4830.T			
4	Signa 4830.TK			
5	Signa 2830.K REPTO			
6	Signa 4225.T (G950)	250HP		
7	Signa 3523.T	5.6L 230 HP		
8	Signa 4623.S			
9	Signa 4023.S 925NM			
10	Signa 2818.TK RMC	5.6L 180HP		
11	Signa 5521.S	5L 210HP		
12	Signa 3521.T (5.57)			
13	Signa 2821.TK RMC			
14	LPK 2821.TK 709 Cab RMC			
15	Signa 2820.T CNG	5.7L CNG		
16	Cowl 2820.T CNG			

Best Regards,
Swapnil Jadhav
PED - HCV
ERC Pune - Vehicle Engg. (CVBU)
Mo. No. - +91 9922001494



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